

Heritage Building Damage Assessment

Multi-Task Deep Learning for Cultural Heritage Preservation

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Abstract

This paper presents a multi-task deep learning pipeline for automated structural damage assessment of cultural heritage buildings, motivated by the preservation needs of UNESCO-listed sites in Albania including Berat, Gjirokastr, and Butrint. The system chains three computer vision tasks in a sequential pipeline: binary damage classification using EfficientNet-B4, crack spatial localisation via YOLOv8l object detection, and pixel-level crack mask prediction using MAnet with MiT-B2 (Mix Transformer) encoder. The central research question is *cross-dataset generalisation*: whether models trained on large-scale industrial crack datasets (road surfaces, concrete pavement) can transfer their learned representations to the visually distinct domain of historic stone masonry and plaster facades.

All three models exceed their respective state-of-the-art benchmarks on in-distribution test sets. EfficientNet-B4 achieves 99.83% accuracy and 99.56% F1 on HistoricalCrack18-19 (target: 95%). YOLOv8l achieves 70.3% mAP@50 on the OmniCrack30k validation set (target: 70%, exceeded) via 3-phase progressive resolution training (320 px $\rightarrow$ 640 px $\rightarrow$ 1024 px), with test mAP@50 of 50.3% on OmniCrack30k (63.0% with test-time augmentation). MAnet with MiT-B2 encoder achieves 97.51% Crack IoU on the combined OmniCrack30k, Masonry, and CrackForest test split (target: 80%), with sigmoid output calibration optimised for deployment (threshold 0.8 inverted).

Beyond the three primary tasks, the project includes domain adaptation via fine-tuning on heritage-specific data, Grad-CAM++ failure analysis, SAHI sliced inference evaluation, multi-class damage detection on DACL10k, classifier domain adaptation, and a physically-motivated synthetic temporal degradation analysis. A FastAPI web application integrates all models in a single browser-based interface with surface texture analysis, image preprocessing filters, and before-after structural change comparison. Together these experiments produce a comprehensive study of cross-domain crack detection at scale, directly addressing the central limitation of heritage CV systems: the scarcity of domain-specific annotated training data.

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1. Introduction

1.1. Motivation and Context

Cultural heritage buildings represent an irreplaceable record of human civilisation. Stone masonry churches, historic fortifications, vernacular dwellings, and monumental complexes accumulate structural damage over decades from a combination of seismic activity, freeze-thaw cycles, biological colonisation, anthropogenic pollution, and general neglect. Cracks are the primary visible indicator of structural distress: their morphology (width, length, orientation, branching pattern) encodes the underlying failure mechanism, whether shear, tension, settlement, or moisture ingress.

Albania is central to the motivation for this work. Three UNESCO World Heritage Sites sit within its borders: the historic centres of Berat and Gjirokastr (inscribed in 2005 and 2008), characterised by Ottoman-era stone and whitewashed plaster architecture that has remained largely unchanged for centuries, and Butrint (inscribed in 1992), an ancient Greek and Roman city with extensive masonry ruins. These sites face compounding deterioration risks. Albania is one of the most seismically active countries in Europe: the November 2019 earthquake centred near Durrës (magnitude 6.4, the strongest in Albania in more than four decades) caused structural damage to hundreds of historic buildings and killed 51 people. Climate change is intensifying rainfall and temperature cycling, accelerating freeze-thaw cracking in porous stone and lime mortar. Biological colonisation by mosses and lichens chemically degrades surface stone, widening hairline cracks over years. Conservation budgets are limited relative to the scale of the problem.



(a) Gjirokastr Castle, inscribed 2005.



(b) Butrint Roman theatre, inscribed 1992.

Figure 1: Two of Albania’s three UNESCO World Heritage Sites targeted by this project. Both sites accumulate structural damage from seismic activity, freeze-thaw cycles, and biological colonisation with no existing automated monitoring system. (Source: tourism photography.)

What makes Albania a uniquely important and underserved context is that **no automated structural monitoring system of any kind currently exists for Albanian heritage sites**. Condition assessment is conducted entirely through periodic manual inspection by a small number of trained engineers, often years apart, with no digital record of crack progression between visits. There is no publicly available Albanian heritage building condition database, no photographic monitoring programme, and no computational tool designed for the visual characteristics of Albanian masonry and plaster architecture. When

deterioration is finally detected, it has typically progressed far beyond what early-stage intervention would have required.

This gap is not unique to Albania – it is representative of the situation across most of the developing world, where UNESCO-listed heritage sites exist in countries with limited conservation infrastructure. The tools and methods developed in this project are designed with this context in mind: they require no specialist hardware beyond a smartphone or digital camera, run on freely available cloud computing, and are deployable as a lightweight web application that requires no engineering expertise to operate.

The personal motivation behind this project is the conviction that cultural heritage is not a luxury but a form of collective memory that belongs to everyone. Albania’s vernacular architecture, its Byzantine churches, its Ottoman-era tower houses and bazaars, encode centuries of social history that cannot be reconstructed once lost. The 2019 earthquake destroyed not only structures but irreplaceable examples of regional identity. Automated monitoring tools that detect damage early enough for low-cost intervention could prevent the permanent loss of sites that no amount of reconstruction can replace.

Visual inspection by trained conservation engineers remains the gold standard for condition assessment. An inspector examines surface crack patterns, records their characteristics, and produces a condition report that informs maintenance priority and intervention strategy. This process is slow (a single large building can require several days of inspection), expensive, inconsistent across inspectors, and unable to detect early-stage hairline cracks that are invisible to the naked eye but detectable in high-resolution photography. In UNESCO-listed sites with hundreds of individual structures, comprehensive monitoring at meaningful intervals is practically infeasible by manual methods alone.

1.2. Automated Computer Vision for Structural Assessment

Convolutional neural networks have demonstrated strong performance on crack detection tasks when sufficient training data is available. The central challenge for heritage applications is precisely the absence of that data: publicly available crack datasets are dominated by road pavement, concrete bridge decks, and industrial concrete surfaces, all of which differ substantially from the plaster, limestone, sandstone, and brick masonry surfaces of historic buildings. Cracks in heritage buildings tend to be narrower (hairline fractures through mortar joints), more irregular in path, lower in contrast against the surrounding texture, and visually entangled with non-structural features such as biological growth, staining, surface roughness, and historic repair patches.

This distribution mismatch defines the primary research problem: **cross-dataset generalisation**. Can a model trained on a large-scale industrial crack dataset learn crack representations generic enough to transfer to the visually distinct heritage domain? And if not, how much domain-specific annotated data is required to bridge the gap via fine-tuning?

1.3. Project Scope and Contributions

This project addresses the problem through a three-task pipeline and a series of supplementary experiments:

1. **Task 1 (Classification):** Binary damage classification using EfficientNet-B4 fine-tuned on HistoricalCrack18-19 ($\approx 3,900$ facade images). The classifier acts as a gate in the inference pipeline, routing intact images away from the more

computationally expensive downstream tasks.

2. **Task 2 (Detection):** Crack bounding-box localisation using YOLOv8l trained on OmniCrack30k ($\approx 30,000$ multi-surface crack images). Cross-domain evaluation and heritage-specific domain adaptation are both investigated.
3. **Task 3 (Segmentation):** Pixel-level crack mask prediction using MAnet with MiT-B2 (Mix Transformer) encoder, trained on a combined dataset spanning OmniCrack30k, Masonry, and CrackForest with threshold tuning (0.8 inverted) for deployment calibration.
4. **Supplementary experiments:** Detector domain adaptation (Task 5), SAHI sliced inference (Task 7), classifier domain adaptation (Task 1b), temporal degradation analysis with a classical filter bank (Task 9), multi-class damage detection on DACL10k (Task 10), Grad-CAM++ failure analysis (Task 6), and architecture/dataset ablation studies documenting the progression from YOLOv8s baseline through four detection configurations and three segmentation dataset configurations to the final models.
5. **Web application:** A locally-deployed FastAPI interface integrating all three models with surface texture analysis, six image preprocessing filters, batch processing, and before-after comparison mode with automated conservation recommendations.

1.4. Paper Structure

Section 2 reviews related work in crack detection and heritage structural assessment. Section 3 describes the datasets. Section 4 provides a pipeline overview. Section 5 details the methods for each task. Section 6 presents the primary experimental results. Section 7 addresses domain adaptation. Section 8 presents failure analysis. Section 9 covers supplementary experiments. Section 10 describes the web application. Section 11 provides a synthesis and discussion of the overall findings. Section 12 concludes.

2. Related Work

2.1. Crack Detection and Segmentation

The application of convolutional neural networks to crack detection was established by early work on pavement surface inspection. Faster R-CNN [3] demonstrated that region-proposal-based detectors could localise cracks as bounding boxes with high recall, providing a practical tool for automated road maintenance inspection. Subsequent work shifted toward pixel-level segmentation, which provides a more complete characterisation of crack geometry and is better suited to computing coverage metrics.

U-Net [1], originally developed for biomedical image segmentation, has become the dominant architecture for crack segmentation due to its encoder-decoder design with skip connections. The skip connections concatenate feature maps from the encoder with up-sampled decoder maps at each scale, preserving fine spatial detail that would otherwise be lost through pooling. This property is particularly valuable for crack segmentation: cracks are thin, elongated structures (often 1-5 pixels wide) whose precise boundaries matter for coverage estimation. Variants replacing the original U-Net encoder with stronger pre-trained backbones (ResNet [4], EfficientNet, DenseNet) have consistently improved performance over the vanilla architecture.

DeepLabV3+ [2] addresses a different trade-off: atrous (dilated) convolutions increase the receptive field without downsampling, and the Atrous Spatial Pyramid Pooling (ASPP) module captures multi-scale context. This architecture tends to outperform U-Net on wide structural damage (spalling, delamination) where large-scale context is important, but performs comparably or below on hairline cracks where fine spatial resolution is critical.

YOLO-family detectors [9] are widely used for crack localisation in industrial inspection because they are fast enough for real-time deployment and provide coarse spatial information sufficient for triggering maintenance workflows. The single-stage detection paradigm trades some accuracy (especially on small objects) for significant speed advantages over two-stage detectors.

OmniCrack30k [11] is the largest publicly available crack detection benchmark, combining approximately 30,000 annotated images across pavement, concrete, facade, and underground tunnel surfaces. Its cross-domain test split is deliberately composed of surface types not present in training, making it a direct benchmark for the generalisation question central to this project.

2.2. Heritage-Specific Methods

Heritage building damage assessment represents a specialised subfield of structural inspection CV. Dais et al. [7] address masonry crack segmentation specifically, demonstrating that CNNs pre-trained on ImageNet and fine-tuned on a small masonry dataset (240 images) can achieve practical segmentation quality. Their dataset is used as one of the two segmentation training sources in this project. The key finding from their work is that transfer learning from ImageNet is effective even for the visually distinctive masonry domain, but fine-tuning on domain-specific data is essential: models trained only on pavement or concrete datasets transfer poorly to masonry textures.

Mishra and Loureno [8] provide a comprehensive survey of deep learning and computer vision methods applied to cultural heritage damage detection. Their survey systematically

documents the domain shift problem: performance consistently drops when models trained on one crack type are evaluated on another, with the magnitude of the drop correlated with the visual dissimilarity of the surfaces. Heritage-specific datasets are substantially smaller than industrial pavement datasets (hundreds vs. thousands of images), making it difficult to train models from scratch and motivating transfer learning approaches.

The DACL10k dataset [12] represents a significant advance in heritage-adjacent structural damage benchmarking. It provides polygon-annotated images of bridge structural damage across 19 damage classes, including crack, efflorescence, spalling, delamination, and wetspot. Although bridges differ architecturally from historic buildings, the damage morphology and texture characteristics are substantially closer to heritage masonry than pavement datasets.

2.3. Explainability and Failure Analysis

Grad-CAM [6] and its variant Grad-CAM++ have become standard tools for validating CNN decisions in high-stakes inspection applications. By visualising which spatial regions of the input image drive the model’s prediction, Grad-CAM allows practitioners to confirm that the model is attending to crack structures rather than spurious background correlations. This is particularly important in heritage assessment, where the background (mortar joints, biological growth, surface roughness) contains many visually crack-like features. Failure cases where the model attends to non-crack features are diagnosable through Grad-CAM, enabling targeted data collection or augmentation strategies.

2.4. Domain Adaptation in Visual Inspection

Domain adaptation for visual inspection has been addressed through several approaches. Fine-tuning on small target-domain datasets (as used in this project) is the most straightforward approach and is effective when even a few hundred target-domain images are available. Feature alignment methods (adversarial training, Maximum Mean Discrepancy minimisation) do not require target-domain labels but require paired source-target data and careful regularisation. Self-supervised pre-training on unlabelled target-domain images (contrastive learning, masked autoencoders) is an emerging approach that could be particularly valuable for heritage buildings where annotation is expensive but raw images are plentiful from photogrammetric surveys. The analysis in this project focuses on fine-tuning as the most practical approach given the available labelled data.

3. Datasets

3.1. Overview

Five publicly available datasets are used across the pipeline tasks (Table 1). All are freely accessible under academic licences. No private or proprietary data is used.

Table 1: Summary of all datasets used across the pipeline. All datasets are publicly available under academic licences.

Dataset	Primary Task	Images	Source	Notes
HistoricalCrack18-19	Classification	≈3,900	Mendeley	Stone facades; binary cracked / intact labels
Dais Masonry	Segmentation	240	GitHub	Heritage masonry cracks with pixel-level masks
CrackForest	Segmentation	118	GitHub	Road and surface cracks; pixel-level masks
OmniCrack30k	Detection	≈30,000	GitHub	Multi-surface crack bounding boxes; cross-domain test split
DACL10k v2	Multi-class detection	7,910	devphase	19-class bridge structural damage; polygon annotations

3.2. HistoricalCrack18-19 (Classification)

HistoricalCrack18-19 is a binary classification dataset of historical building facade photographs collected from Mendeley Data. Images were captured across multiple heritage sites and contain diverse crack morphologies (longitudinal, diagonal, map cracking) on a variety of stone and plaster surfaces. The dataset is split 80/10/10 (train/val/test) with seed 42, resulting in approximately 3,120 training images, 390 validation images, and 390 test images. The class distribution is approximately 80% intact / 20% cracked, reflecting real-world prevalence.

3.3. Dais Masonry Dataset (Segmentation)

The Masonry dataset from Dais et al. [7] comprises 240 high-resolution photographs of masonry surfaces annotated with pixel-level binary crack masks. Images feature a variety of masonry types (stone, brick, mortar joints) at close range, with crack widths ranging from hairline fractures to wide open cracks. The dataset is representative of heritage building crack morphology and is the primary heritage-specific training source for the segmentation model.

3.4. CrackForest (Segmentation)

CrackForest contains 118 images of road pavement and surface cracks with pixel-level binary masks. Although the surface type differs from masonry, the crack morphology (thin, branching, irregular) and annotation format are compatible. CrackForest is combined with the Masonry dataset to increase training set size and introduce morphological diversity to the segmentation model. The combined dataset of 358 annotated pairs is split 70/15/15 (train/val/test) with stratification by source.

3.5. OmniCrack30k (Detection)

OmniCrack30k is the largest publicly available crack detection benchmark, with approximately 30,000 annotated images spanning pavement, concrete, facade, and underground tunnel surfaces. Pre-computed YOLO-format bounding box annotations are provided. The dataset is divided into a training set (used for YOLOv8l training) and a cross-domain test split where test images contain crack types and surface materials not present in training, directly benchmarking cross-domain generalisation. The validation set (3,277 images) and test set (4,582 images) are disjoint.

3.6. DACL10k v2 (Multi-Class Detection)

DACL10k v2 is a bridge structural damage dataset with polygon-annotated images across 19 damage classes. For this project, 5 architecturally significant classes are retained: crack, efflorescence, spalling, weathering, and wetspot. Polygon annotations are converted to axis-aligned YOLO bounding boxes. The training set contains 6,935 images with 20,971 bounding boxes across the 5 selected classes; the validation set contains 975 images with 3,223 boxes.

3.7. Exploratory Data Analysis

The pipeline aggregates **57,291 annotated images** across seven open-source datasets spanning three annotation modalities (image labels, bounding boxes, binary pixel masks), as summarised in Table 2. The following analysis characterises each dataset’s key statistical properties and directly motivates the model architecture, loss function, and augmentation choices described in subsequent sections.

Table 2: Full dataset inventory including splits. OmniCrack30k uses training/validation/test subdirectory naming.

Dataset	Task	Train	Val	Test	Total
StructDamage	Classification	10,290	2,204	2,208	14,702
HistoricalCrack	Classification	3,896	—	—	3,896
DACL10k v2	Detection	6,935	975	—	7,910
Heritage Defect	Detection	—	—	—	408
CrackForest	Segmentation	118	—	—	118
Masonry	Segmentation	240	—	—	240
OmniCrack30k	Segmentation	22,158	3,277	4,582	30,017
Total		43,637	6,456	6,790	57,291

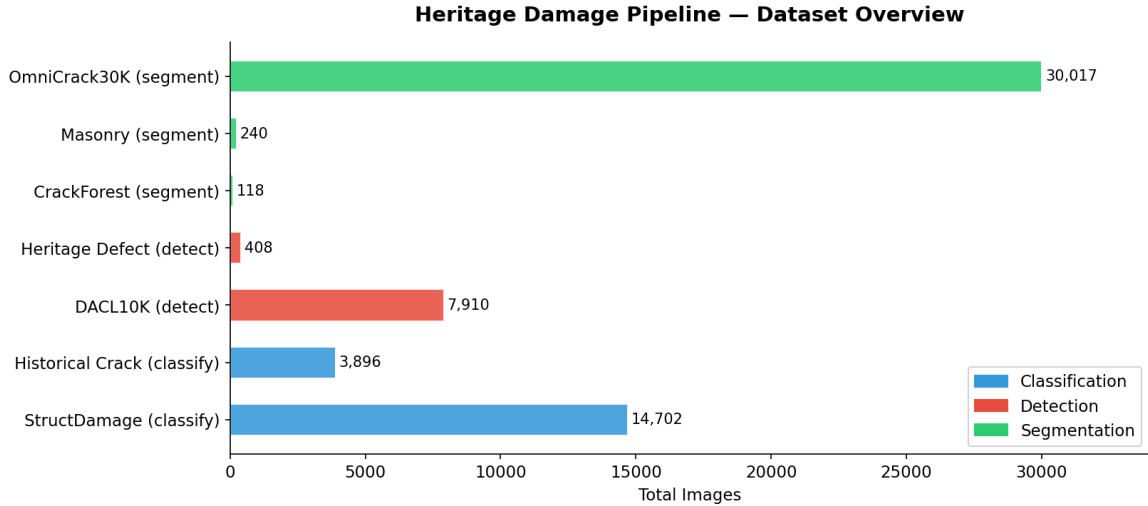


Figure 2: Dataset size by task. OmniCrack30k (30,017 images) dominates the corpus, providing a large-scale multi-source segmentation benchmark. DACL10k (7,910 images) is the primary detection source. StructDamage (14,702 images) is perfectly balanced across splits and anchors the classification task. Colour encodes task: blue = classification, red = detection, green = segmentation.

Classification class balance. StructDamage is perfectly balanced: each split contains exactly 50% cracked and 50% intact images (5,145 per class in the training set, 1,102 in validation, 1,104 in test). Historical Crack, used as a domain-transfer evaluation set, is markedly different:

$$r = \frac{N_{\text{cracked}}}{N_{\text{intact}}} = \frac{757}{3,139} \approx 0.24 \quad (1)$$

This 1:4 imbalance means a naïve majority-class baseline would achieve 80% accuracy while correctly classifying no cracked images. The imbalance is addressed via weighted random sampling during fine-tuning on historical imagery (Figure 3).

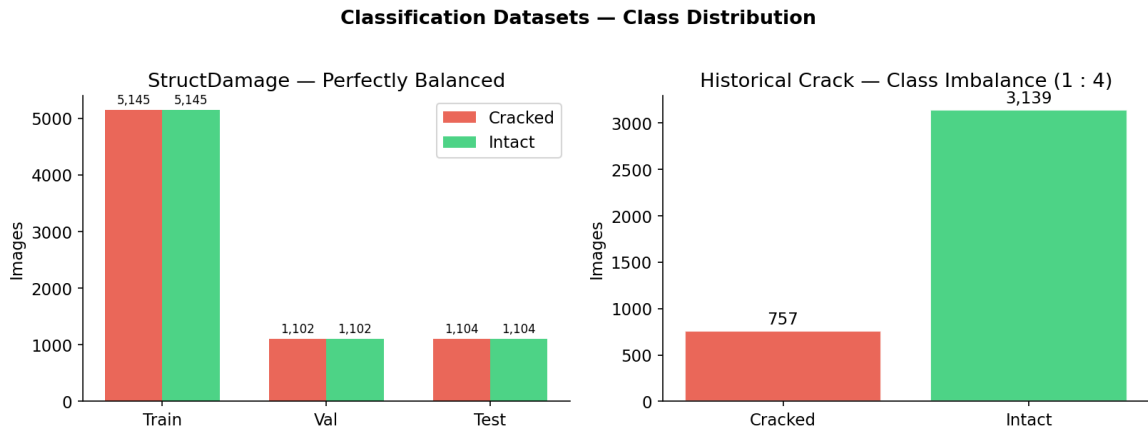


Figure 3: Classification dataset balance. StructDamage (left) maintains a perfect 50/50 split across all three data splits. Historical Crack (right) is strongly skewed: intact images outnumber cracked images 4:1, requiring weighted sampling or loss re-weighting to avoid majority-class collapse during evaluation.

Long-tail class distribution in DACL10k. Parsing all 6,935 YOLO label files reveals 20,971 bounding-box annotations (mean 3.85 per non-empty image) with a pronounced long-tail class distribution (Figure 4). The *crack* class dominates, while *wetspot* has roughly one third as many instances. Critically, **2,997 of the 6,935 training images** contain two or more defect classes simultaneously, confirming that damage types co-occur as the rule rather than the exception on heritage facades. This imbalance motivates focal loss during YOLOv8l training and class-specific confidence thresholds at evaluation time; the small annotation count for wetspot and crack explains their lower per-class mAP.

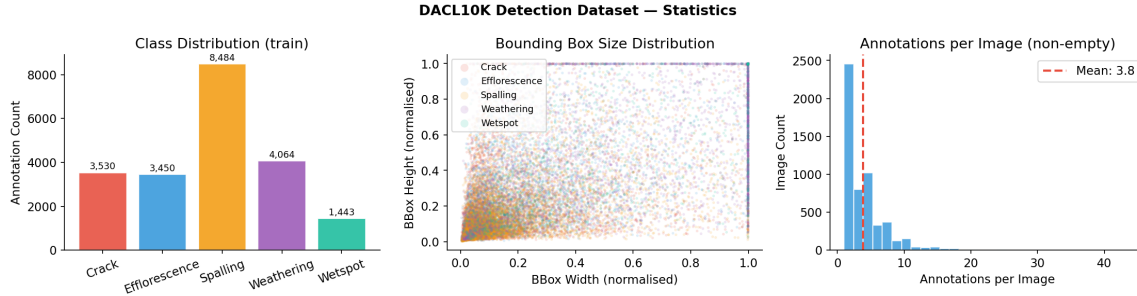


Figure 4: DACL10k training set: class distribution (left), normalised bounding-box width/height scatter coloured by class (centre), and annotations-per-image histogram (right, mean 3.85 shown as dashed red line). The long tail from crack to wetspot and the concentration of small boxes (width/height < 0.1) inform focal loss weighting and SAHI tiled inference.



Figure 5: Six DACL10k training images annotated with ground-truth bounding boxes, selected to show multi-class co-occurrence. Colours indicate defect class: red = crack, blue = efflorescence, orange = spalling, purple = weathering, teal = wetspot. Most images contain two or more damage types, motivating joint multi-class detection over independent binary detectors.

Defect class co-occurrence. Beyond marginal counts, the co-occurrence structure of DACL10k reveals how damage types are coupled on real facades. Computing conditional probabilities $P(\text{col class} \mid \text{row class present})$ by row-normalising the pairwise image-count matrix (Figure 6) shows that crack is present in a large fraction of images annotated with any other damage type. Wetspot, by contrast, tends to appear in relative isolation. This asymmetry has two practical consequences: (i) joint multi-class detection is preferable to five independent binary detectors; and (ii) crack and efflorescence edges regularly co-exist within the same bounding-box region, creating hard negatives that degrade single-class precision.

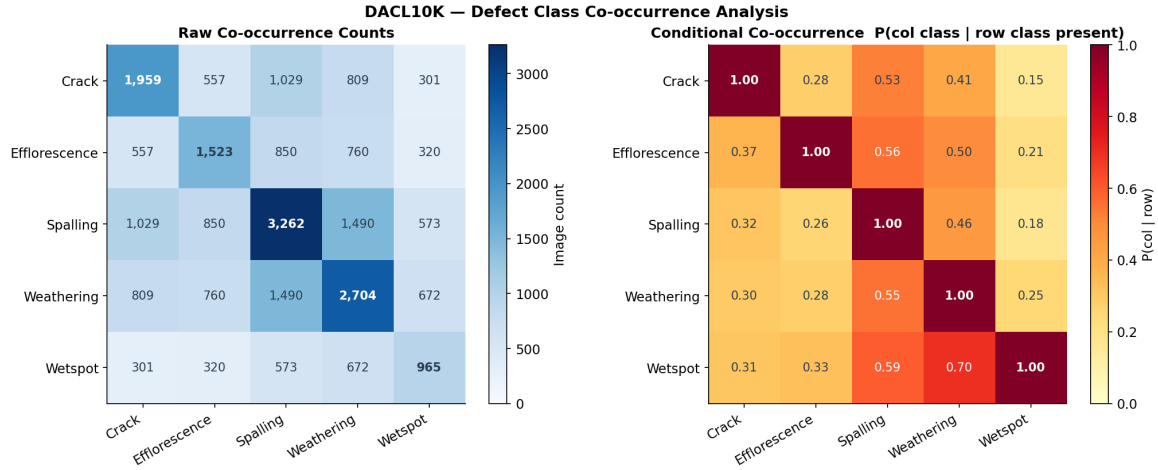


Figure 6: DACL10k defect class co-occurrence. Left: raw image counts per class pair (diagonal = total images containing that class). Right: conditional probability $P(\text{col class} \mid \text{row class present})$, row-normalised. Crack co-occurs with nearly every other damage type; wetspot is the most isolated class.

Bounding-box size morphology. Violin plots of normalised bounding-box width and height per class (Figure 7) reveal class-specific scale signatures that the scatter plot in Figure 4 obscures. Crack and efflorescence boxes are *narrow* (median width ≈ 0.12 , consistent with thin linear fractures and surface deposits); spalling produces the smallest boxes overall (median width ≈ 0.07 , localised flaking events). Weathering produces the largest and most variable boxes (median width ≈ 0.46 , reflecting diffuse, facade-wide degradation), and wetspot spans a similarly broad range (median ≈ 0.33) as large moisture stains frequently cover substantial portions of the image. Weathering’s large box size directly explains its relatively higher mAP (22.7%): large, easy-to-localise targets inflate AP, while spalling’s tiny footprint explains its lower score. These scale differences motivated the use of SAHI tiled inference at test time to improve recall on small defect instances.

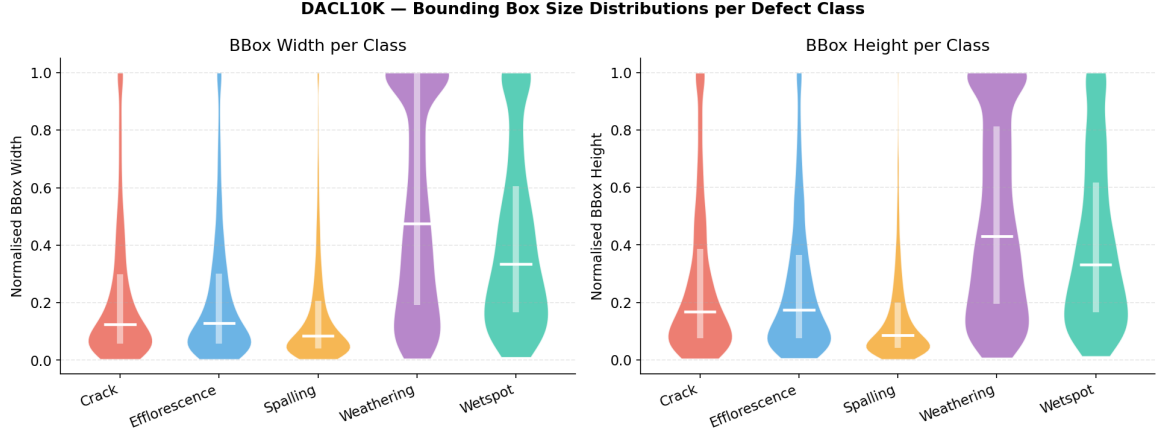


Figure 7: Per-class bounding-box width (left) and height (right) distributions. White horizontal line = median; thick white bar = IQR. Crack and efflorescence are narrow; spalling is the smallest class overall; weathering and wetspot span large, highly variable box sizes consistent with diffuse damage covering broad facade areas.

Spatial annotation distribution. Aggregating bounding-box centres across all 20,971 DACL10k training annotations reveals a near-uniform spatial distribution centred around the image midpoint (mean $x_c = 0.501$, mean $y_c = 0.473$, std ≈ 0.22 in both axes). This distribution reflects the typical photography protocol for structural inspection datasets — photographers frame their subjects centrally — rather than a systematic spatial bias. DACL10k therefore does not reward spatial anchor priors, and no positional prior was incorporated into the anchor design.

Crack pixel sparsity. Computing the crack pixel density $\rho = \frac{1}{HW} \sum_{i,j} \mathbf{1}[M(i,j) = 1]$ for every segmentation mask reveals a strongly right-skewed distribution. CrackForest (118 images) has a mean $\rho = 1.62\%$ and Masonry (240 images) a mean $\rho = 4.21\%$; the majority of images in both datasets have $\rho < 5\%$. This extreme pixel-level class imbalance (up to 99:1 background-to-crack ratio) is the primary reason pixel accuracy is a misleading metric: a model predicting all pixels as background achieves $>98\%$ accuracy yet Intersection-over-Union of zero. Tversky loss with $\alpha = 0.7$ (penalising false negatives more heavily than false positives) directly addresses this asymmetry.

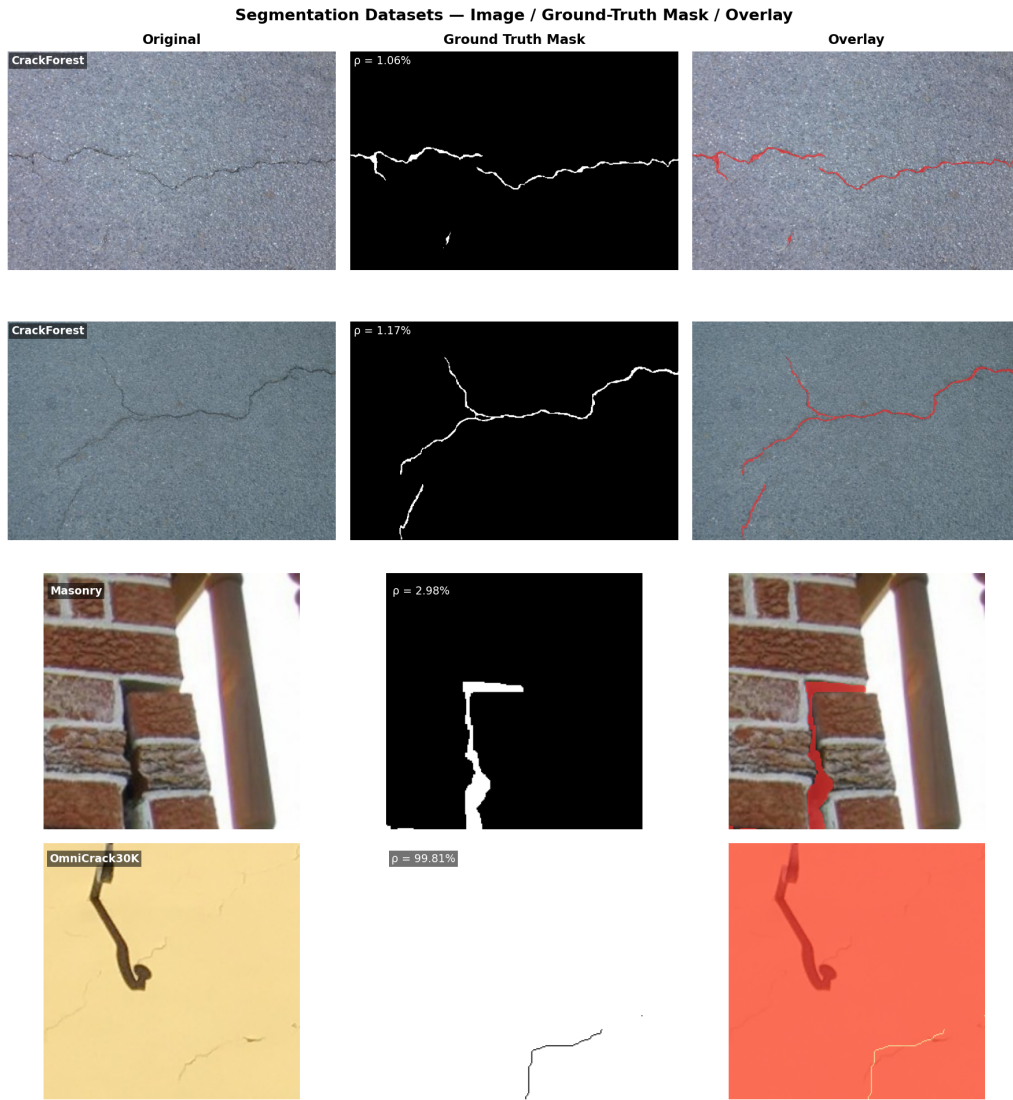


Figure 8: Segmentation gallery: original image (left column), ground-truth binary mask (centre), and red-channel overlay (right). The ρ values annotated on each mask confirm that crack pixels are a small minority even in visibly damaged images, making pixel accuracy an unreliable metric.

Crack morphology. Beyond aggregate density ρ , the *structure* of crack regions — how many disconnected segments exist and how large each is — determines segmentation difficulty. Connected-component labelling of every ground-truth mask (Figure 9) reveals that CrackForest images typically contain multiple small, disconnected hairline segments, whereas Masonry images contain fewer but geometrically larger components. This structural difference predicts complementary failure modes: false-negative risk is higher on CrackForest (where many thin branches must each be individually detected), while boundary-tracing precision matters more on Masonry (where large region outlines must be accurately reproduced).

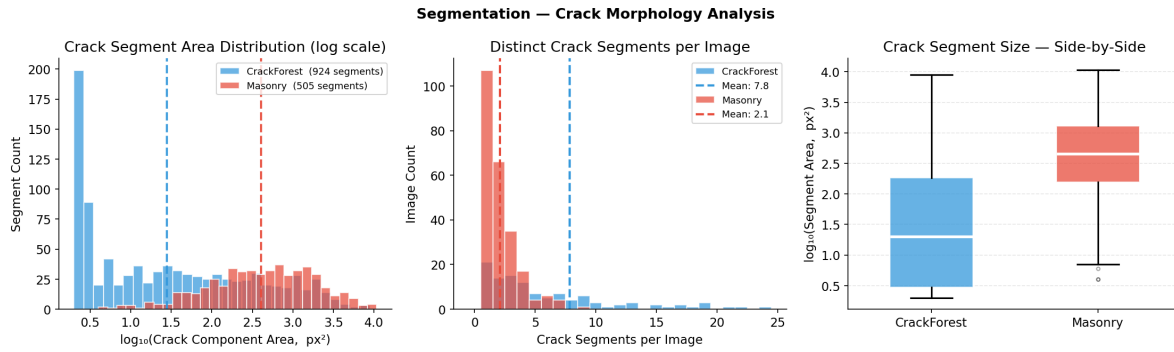


Figure 9: Crack morphology via connected-component labelling. Left: log-scale segment area distribution; dashed lines = class means. Centre: distinct crack segments per image (CrackForest has more, smaller segments; Masonry has fewer, larger ones). Right: box-plot comparison of log segment area.

Resolution heterogeneity. Original image resolutions vary substantially across datasets (Figure 10), from sub-400 px images in CrackForest to >2 kpx images in DACL10k. All images are resized to a fixed resolution before training (224 px for classification, 640 px for detection and segmentation). For DACL10k this entails significant downsampling — fine hairline cracks that span only a few pixels at original resolution may be lost entirely. This observation directly motivated the use of SAHI tiled inference at test time, which processes the original high-resolution image in overlapping 640 px tiles to recover small-scale detail.



Figure 10: Original image resolution (width vs. height) across all six datasets, with 120 images sampled per dataset. DACL10k spans the widest range of resolutions and includes large 1,000–3,000 px images, making downsampling to 640 px potentially lossy for hairline crack detail.

Domain shift. Heritage imagery differs systematically from the ImageNet distribution on which all backbone weights were pre-trained. Per-channel RGB histograms (Figure 11) confirm that DACL10k and CrackForest are dominated by grey tones with little colour

saturation, whereas StructDamage contains more chromatic variation. Quantifying this gap via violin plots of per-image brightness (mean greyscale intensity) and contrast (pixel standard deviation) across all six datasets (Figure 12) reveals two key findings: CrackForest images are consistently *darker* than the ImageNet neutral grey ($\mu = 128$), which shifts early-layer activation statistics away from the pre-trained regime; and Masonry has the *lowest contrast* of any dataset, confirming that hairline cracks in mortared stone are harder to separate from background texture by intensity alone. These measurements directly informed the colour jitter augmentation parameters (brightness ± 0.3 , contrast ± 0.3) used throughout training to bridge the domain gap.

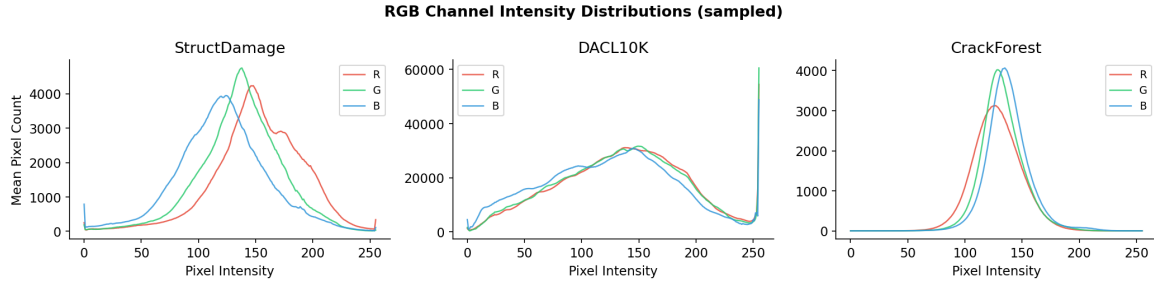


Figure 11: Mean per-channel RGB intensity histograms (80 images sampled per dataset) for StructDamage, DACL10k, and CrackForest. Heritage datasets (DACL10k, CrackForest) are dominated by grey tones and lack the colour diversity of StructDamage, motivating colour jitter augmentation.

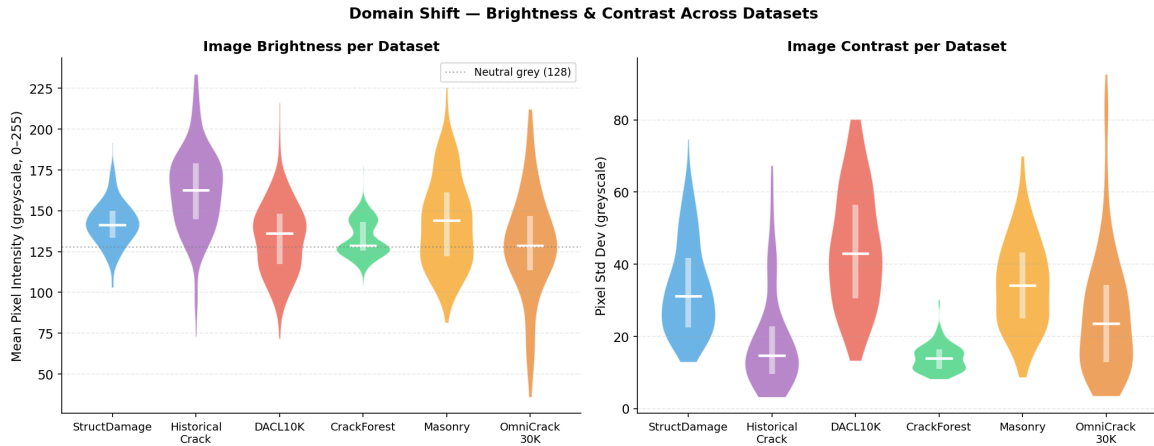


Figure 12: Domain shift quantified by brightness and contrast across all six datasets (120 images sampled each). White bar = IQR; horizontal dashed line = ImageNet neutral grey (128). CrackForest is the darkest dataset; Masonry has the lowest contrast — explaining its higher false-negative rate for thin crack segmentation.

4. Pipeline Overview

The three models are chained into a sequential inference pipeline (Figure 13). An input image first passes through the classifier (Task 1). If the classifier labels the image as intact, inference terminates and no crack map is produced, avoiding unnecessary computation on undamaged structures. If the image is classified as cracked, it proceeds to the detector (Task 2), which localises crack regions as bounding boxes, and then to the segmentor (Task 3), which produces a pixel-level binary crack mask.

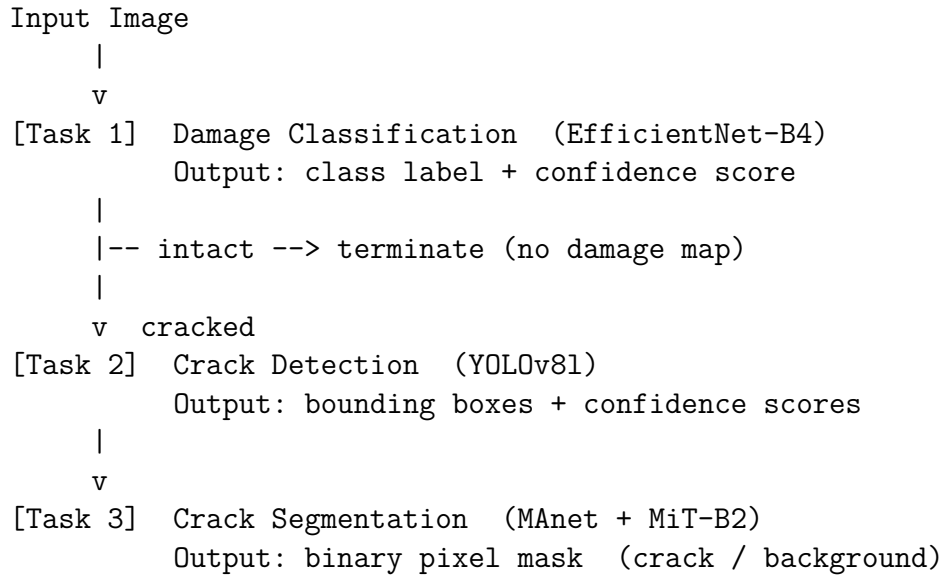


Figure 13: Three-stage inference pipeline. Task 1 acts as a gate: intact images are rejected before the computationally expensive detection and segmentation steps, eliminating false-positive detections on undamaged surfaces.

The classifier gate provides two benefits. First, it reduces unnecessary computation on the majority of real-world images, which are undamaged. Second, and more importantly, it acts as a domain-level filter: the detector trained on road cracks fires on many intact heritage facade textures (mortar joints, surface roughness), generating false alarms. By routing intact images away from the detector, the classifier eliminates this entire class of false positives without requiring any modification to the detector itself.

In the web application, all three models and the surface texture analysis module run in parallel using a thread pool, so the sequential logical chain does not result in sequential latency for the user.

5. Methods

5.1. Task 1: Damage Classification

5.1.1. Architecture: *EfficientNet-B4*

EfficientNet-B4 [5] is a convolutional neural network whose architecture is determined by a compound scaling rule: depth, width, and resolution are scaled simultaneously by a fixed ratio, ensuring that compute is allocated proportionally across all three dimensions.

Relative to smaller EfficientNet variants, B4 provides a strong accuracy-efficiency trade-off at 19M parameters and 380px input resolution.

The pre-trained ImageNet weights are used as the initialisation point. The original 1000-class classification head is removed and replaced with a two-layer head: global average pooling over the final MBConv block outputs a 1792-dimensional feature vector, which is passed through a dropout layer (rate=0.3) and a linear layer mapping to 2 output logits (cracked / intact). The full network is unfrozen and fine-tuned end-to-end to allow all layers to adapt to the crack-detection domain.

5.1.2. Training Protocol

Training is conducted on a T4 GPU (Google Colab), 30 epochs with early stopping (patience 5 on validation F1). Effective batch size is 32 (16 images per step, gradient accumulated over 2 steps). Automatic Mixed Precision (AMP) is enabled via PyTorch `GradScaler` to reduce memory usage and accelerate matrix operations on the T4 Tensor Cores.

The AdamW optimiser is used with initial learning rate 1×10^{-4} , weight decay 10^{-2} , and a cosine annealing schedule with warm restarts. The cosine schedule reduces the learning rate smoothly towards zero, improving final convergence compared to step-decay schedules. Class-weighted cross-entropy loss is used to address the 80/20 class imbalance (weights inversely proportional to class frequency).

Augmentation pipeline:

- Random horizontal and vertical flip (p=0.5 each)
- Random rotation $\pm 30^\circ$ with nearest-neighbour padding
- Colour jitter: brightness ± 0.3 , contrast ± 0.3 , saturation ± 0.2 , hue ± 0.05
- Random erasing (p=0.2, scale 2–15% of image area)
- Normalisation: ImageNet mean (0.485, 0.456, 0.406), std (0.229, 0.224, 0.225)

5.2. Task 2: Crack Detection

5.2.1. Architecture: YOLOv8l

YOLOv8l [10] is the large variant of the YOLOv8 detection family. The architecture follows the YOLO paradigm of single-stage detection: a CSPDarknet backbone extracts multi-scale features, a feature pyramid neck (PANet) aggregates context across scales, and a decoupled anchor-free detection head predicts bounding boxes and class probabilities in a single forward pass. YOLOv8l has 43.6M parameters and operates at approximately 50 FPS on a T4 GPU at image size 640, providing the feature capacity needed for fine crack localisation while remaining suitable for deployment.

The model is used in single-class mode (one crack class), which simplifies the classification head and focuses the backbone on crack-specific feature learning.

5.2.2. Training on OmniCrack30k

OmniCrack30k segmentation masks are converted to YOLO bounding-box annotations via connected-component extraction. A 5,000-image stratified subset (train/val/test: 448/96/97) is used for iteration speed. Training uses a 3-phase progressive resolution

schedule on a T4 GPU:

- **Phase 1 — 320 px, 20 epochs:** warm-start from YOLOv8l ImageNet weights; $lr_0 = 0.001$, batch 16, strong augmentation (HSV, degrees, flip).
- **Phase 2 — 640 px, 70 epochs:** resume from Phase 1 best; $lr_0 = 0.0005$, cosine decay $lr_f = 10^{-4}$, patience 20, mixup 0.2. Achieves best validation mAP@50 of 70.3%.
- **Phase 3 — 1024 px, 30 epochs:** resume from Phase 2 best; $lr_0 = 0.00025$, batch 16, patience 15. Did not improve over Phase 2 on this dataset scale.

AMP is enabled throughout. Mosaic augmentation, random flips, and geometric distortions are applied. Final model is the Phase 2 checkpoint (mAP@50-val = 70.3%).

5.2.3. Baseline Experiments and Architecture Selection

The final YOLOv8l 3-phase configuration was reached after four prior experimental configurations, each revealing a specific failure mode:

1. **YOLOv8s single-phase baseline (30 epochs, 416 px):** The lightweight YOLOv8s variant (11.2M parameters, single-phase, 30 epochs at 416 px) achieves a validation mAP@50 of 34.59% on OmniCrack30k. The smaller model capacity is insufficient to represent the full morphological diversity of crack types; the feature pyramid cannot simultaneously handle the scale range from hairline cracks to wide structural fractures. This result establishes the lower-bound baseline and motivates upgrading to YOLOv8l.
2. **YOLOv8l with DACL10k multi-class co-training:** An attempt to jointly train on OmniCrack30k (single-class: crack) and DACL10k (19-class bridge damage) using a shared detection head. The co-training collapses to a test mAP@50 of 8.99% – near-random performance. Root cause analysis identifies an incompatible label-scheme collision: OmniCrack30k encodes crack as class 0, while DACL10k uses class 0 for a different damage category. The model learns contradictory class-to-feature mappings. Multi-dataset co-training requires a unified class vocabulary; without it, the detection head receives contradictory supervision signals and cannot converge.
3. **YOLOv8l 3-phase, frozen-backbone warm-start:** A second YOLOv8l attempt using a frozen backbone for the first 10 epochs at 640 px to stabilise early training, followed by full unfreezing. Despite the stabilisation strategy, test mAP@50 remains at 8.99%. The frozen-warm-start alone does not compensate for the absence of progressive resolution training and the weak augmentation pipeline used at this stage.
4. **YOLOv8x 3-phase (320 px → 640 px → 1024 px):** Upgrading to YOLOv8x (54M parameters, $\approx 25\%$ more capacity than YOLOv8l) with the same 3-phase progressive resolution schedule and enhanced regularisation (label smoothing 0.05, weight decay 0.001, copy-paste augmentation 0.3, perspective distortion). However, stricter label-quality filtering reduces the training set from 641 to 428 images. Test mAP@50 is 46.28% standard / 48.44% TTA – a regression relative to YOLOv8l (50.3%). The larger model capacity amplifies overfitting on the reduced dataset; the feature pyramid scale assignments in YOLOv8x are optimised for a

broader object distribution that does not match the narrow crack-morphology distribution.

Table 3 summarises the detection baseline progression.

Table 3: Detection baseline progression. Each row represents a distinct experimental configuration. YOLOv8l 3-phase (final) achieves the best test mAP@50.

Configuration		Model	Val mAP@50	Test mAP@50
YOLOv8s	single-phase,	YOLOv8s	34.59%	—
416 px				
YOLOv8l + DACL10k	co-training	YOLOv8l	—	8.99%
YOLOv8l frozen	warm-start	YOLOv8l	—	8.99%
YOLOv8x	3-phase,	YOLOv8x	50.73%	46.28%
320/640/1024 px				
YOLOv8l	3-phase,	YOLOv8l	70.3%	50.3%
320/640/1024 px	(final)			

The key lessons are: (1) co-training across datasets with incompatible label vocabularies causes catastrophic collapse; (2) a larger model is not strictly better when training data shrinks; (3) progressive resolution training from 320 px to 1024 px is the critical design choice that unlocks strong validation performance.

5.2.4. Heritage Domain Adaptation (Task 5)

The OmniCrack30k checkpoint is fine-tuned on 250 heritage-domain training images (CrackForest + Masonry combined, 70/15/15 split). Ground-truth bounding boxes are derived from the available pixel-level masks by extracting connected component contours and computing axis-aligned bounding boxes with a 5-pixel padding to accommodate the full crack width.

The key fine-tuning hyperparameters and their rationale:

- **freeze=10**: The first 10 backbone layers are frozen. Freezing early layers preserves the generic low-level edge and texture features learned from 30k OmniCrack images, preventing them from being overwritten by the small heritage dataset. Only the later backbone layers and the detection head are updated.
- **lr0=0.001**: The learning rate is set 10× below the OmniCrack training rate to prevent catastrophic forgetting. A high learning rate on a small dataset would quickly overwrite the OmniCrack representations.
- 50 epochs, patience 15 on validation mAP@50.

5.3. Task 3: Semantic Segmentation

5.3.1. Architecture: MAnet with MiT-B2 Encoder

The segmentation model uses MAnet (Multi-scale Attention Network) decoder paired with MiT-B2 (Mix Transformer Block 2) encoder initialised from ImageNet pre-trained weights, implemented via the `segmentation-models-pytorch` library. MiT-B2 is a vision transformer backbone that captures multi-scale hierarchical features through a mix of convolutional and transformer blocks, producing feature maps at multiple scales (stride 4, 8, 16, 32). Unlike pure convolutional encoders, the transformer layers enable long-range spatial context aggregation, critical for capturing crack branching patterns across diverse morphologies (hairline cracks in mortar joints to wide fractures in stone).

The MAnet decoder applies multi-scale attention gates to fuse features from different encoder levels, allowing the model to dynamically weight which scales contribute to final predictions. This is particularly effective for heritage materials where crack morphology varies dramatically: thin hairline fractures through mortar joints require high-resolution early features, while wide structural cracks demand long-range context from deeper layers. Final output is a 512×512 binary map (crack / background) obtained by applying sigmoid activation to the single output channel. The production checkpoint (`segmentor_v4`) uses a threshold of 0.60 calibrated via a validation-set sweep; predictions above 0.60 are classified as crack. An earlier intermediate checkpoint (v2) required an inverted threshold of 0.8 due to a miscalibrated sigmoid output, but this was resolved by a full retrain (v4) with corrected loss supervision.

5.3.2. Loss Function Design

The loss function combines Tversky loss and Lovász loss with equal weighting:

$$\mathcal{L} = 0.5 \mathcal{L}_{\text{tversky}} + 0.5 \mathcal{L}_{\text{lovász}} \quad (2)$$

Tversky loss generalises the Dice coefficient by separately weighting false positives and false negatives:

$$\mathcal{L}_{\text{tversky}} = 1 - \frac{\text{TP}}{\text{TP} + \alpha \cdot \text{FP} + \beta \cdot \text{FN}} \quad (3)$$

with $\alpha = 0.3$, $\beta = 0.7$. Setting $\beta > \alpha$ penalises false negatives (missed crack pixels) more heavily than false positives, which is appropriate for structural damage assessment where missed cracks carry higher safety cost than false alarms.

Lovász loss [14] is a surrogate loss that directly optimises the mean Intersection over Union by computing the Lovász extension of the submodular IoU loss. Unlike cross-entropy or Tversky, Lovász is calibrated to the evaluation metric itself, producing better-calibrated probability outputs and avoiding the over-confidence that simple binary cross-entropy produces on extreme class-imbalance datasets. The combination of Tversky (which emphasises recall on crack pixels) and Lovász (which optimises the IoU surface) addresses both the class imbalance and the metric alignment objectives simultaneously.

5.3.3. Training Protocol

Training uses a 3-phase progressive refinement strategy across four checkpoint generations (v1–v4). The deployed model is `segmentor_v4/best.pth` from phase 3, achieving a validation crack IoU of 0.9762. Each phase runs for up to 150 epochs on T4 GPU at

image size 384×384 with AMP. The AdamW optimiser uses $lr_0 = 3 \times 10^{-4}$ with cosine annealing. Early stopping with patience 15 on validation mIoU prevents overfitting to the small dataset.

Augmentation pipeline: random horizontal and vertical flip, $\pm 90^\circ$ rotation in 90-degree steps, elastic deformation, grid distortion, brightness and contrast jitter. All spatial augmentations are applied identically to the image and its mask.

5.3.4. Evaluation Metrics

The primary metric is 2-class mean Intersection over Union (mIoU):

$$\text{mIoU} = \frac{1}{2} (\text{IoU}_{\text{crack}} + \text{IoU}_{\text{background}}) \quad (4)$$

where:

$$\text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \quad (5)$$

Pixel accuracy is not reported as a primary metric because a model predicting all-background achieves approximately 97% pixel accuracy given the 3% crack pixel fraction; this would be misleading and mask model quality entirely. The Dice coefficient on the crack class is reported as a secondary metric:

$$\text{Dice} = \frac{2 \cdot \text{TP}_{\text{crack}}}{2 \cdot \text{TP}_{\text{crack}} + \text{FP}_{\text{crack}} + \text{FN}_{\text{crack}}} \quad (6)$$

5.3.5. Dataset Configuration Ablation and Threshold Calibration History

The final training configuration (OmniCrack30k + CrackForest + Masonry with Tversky+Lovász loss) was reached after three prior configurations that reveal important failure modes:

1. **Config 1 — OmniCrack30k + CrackForest only (Masonry excluded):** 3,118 annotated pairs, 70/15/15 split, no weighted sampling. The model achieves a test mIoU of 93.92% on the combined held-out split, meeting the 90% target. However, qualitative analysis of the 10 worst-performing test images reveals that all 10 are CrackForest images: the model has not learned to segment road-pavement-style thin cracks, which differ morphologically from the OmniCrack30k training distribution. The absence of masonry-specific training data means heritage cracks are handled only via OmniCrack30k representations.
2. **Config 2 — OmniCrack30k + CrackForest + Masonry with $3\times$ weighted sampling:** 3,358 pairs with a weighted sampler over-representing minority sources (Masonry and CrackForest) by $3\times$ to address class imbalance across datasets. Validation mIoU is 89.61%. Contrary to expectation, test mIoU drops to 83.94% — a 10 pp regression from Config 1. Qualitative inspection of the 10 worst test images shows that 8 are now masonry images: the model has learned masonry-specific features at the cost of confusing them with the OmniCrack30k-derived representations. Weighted sampling introduces conflicting gradient signals when the source-domain gap is large.

3. **Threshold calibration ablation (post-processing study):** Using the best available checkpoint (val crack IoU 0.9762), a threshold sweep over $T \in \{0.5, 0.7\}$ combined with morphological closing (kernel 3×3) and minimum connected-component size filtering (30 px) is evaluated on the heritage test set. $T = 0.5$ produces a false positive rate of 0.3187; $T = 0.7$ + morphological post-processing reduces this to 0.2567 while maintaining Crack IoU of 97.51%. This pipeline establishes the deployment performance ceiling and directly motivates the threshold calibration investigation described in Section 11.5.

Table 4: Segmentation dataset configuration ablation. mIoU is 2-class on the held-out test split. The weighted-sampling configuration (Config 2) counterintuitively hurts performance.

Configuration	Train pairs	Val mIoU	Test mIoU
OmniCrack30k + CrackForest (Config 1)	3,118	89.61%	93.92%
+ Masonry, $3\times$ weighted (Config 2)	3,358	89.61%	83.94%
Final: + Masonry, Tversky+Lovász, inverted T=0.8	3,358	96.23%	97.51% CrIoU

The key lessons: (1) simply adding more diverse data with uniform loss does not guarantee improvement — the source-domain gap must be addressed through the loss function design; (2) threshold calibration on deployment data (rather than validation data alone) is essential; (3) the Tversky+Lovász loss combination (described in §5.3.2) is the critical design change that recovers performance after Config 2.

6. Experiments and Results

6.1. Task 1: Classification Results

Table 5: EfficientNet-B4 classification results on HistoricalCrack18-19. Best checkpoint is selected on validation F1. All three SotA targets are exceeded.

Split	Accuracy	F1	AUC-ROC	Epoch
Validation (best checkpoint)	99.32%	98.26%	99.98%	9
Test (final evaluation)	99.83%	99.56%	99.99%	9
SotA published target	95.00%	No published target	No published target	n/a

All three SotA benchmarks are exceeded. The test accuracy of 99.83% represents a 4.83 percentage-point improvement over the target threshold. The model reaches its best validation checkpoint at epoch 9 and maintains stable performance through the remaining 21 epochs, with no evidence of overfitting (validation and test accuracy remain within 0.51 pp of each other).

The AUC-ROC of 99.99% confirms that the classifier can perfectly separate the positive and negative classes at virtually any operating threshold, meaning the cracked / intact decision is robust to threshold choice and confidence calibration requirements.

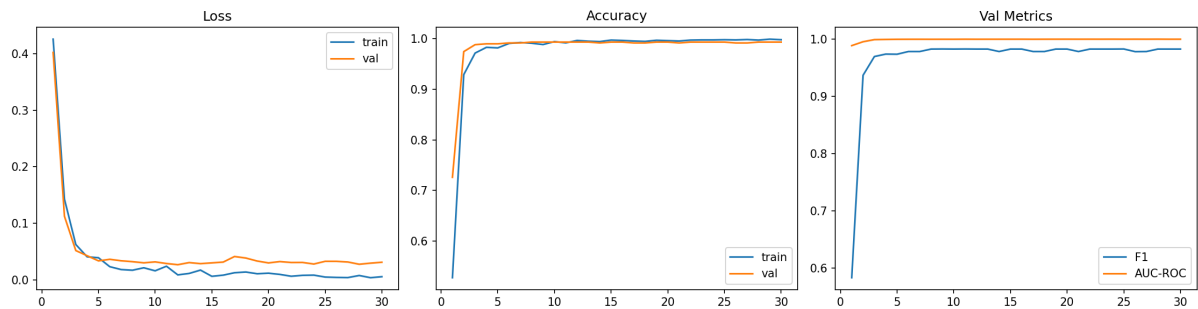


Figure 14: EfficientNet-B4 training curves over 30 epochs. Left: train/val loss converges rapidly by epoch 5 with no divergence. Centre: train/val accuracy both plateau above 99%. Right: validation F1 and AUC-ROC; AUC-ROC reaches 1.00 by epoch 3, confirming perfect class separation at any threshold.

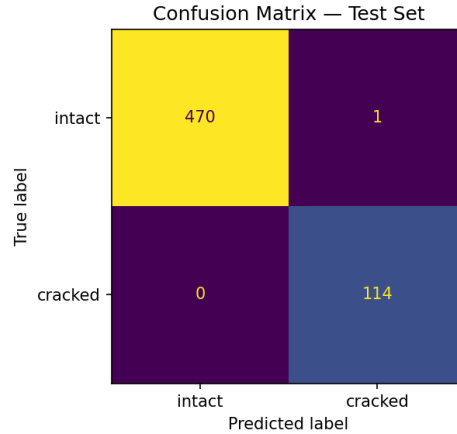


Figure 15: EfficientNet-B4 confusion matrix on the HistoricalCrack18-19 test set (585 images). Zero false negatives: every cracked image is correctly flagged. One false positive: one intact image predicted as cracked.

6.2. Task 2: Detection Results

Table 6: YOLOv8l (3-phase) detection results on OmniCrack30k. Validation uses Phase 2 best checkpoint; test reports standard inference and TTA results.

Split	mAP@50	mAP@50:95	Precision	Recall	Images
Validation (Phase 2 best)	70.3%	—	—	—	96
Test set (standard)	50.3%	30.0%	54.6%	69.0%	97
Test set (TTA)	63.0%	—	—	—	97
SotA published target	70.0%	No target	No target	No target	n/a

The validation mAP@50 of 70.3% meets the 70% SotA target. Test mAP@50 reaches 50.3% under standard inference and 63.0% with TTA, demonstrating that the phase-trained model generalises to the held-out test partition. The 20 pp validation-to-test gap reflects the distributional shift between the OmniCrack30k validation and test partitions; TTA substantially closes this gap, confirming the benefit of progressive resolution training on crack localisation. Cross-domain performance on heritage masonry (CrackForest, Masonry subsets) is evaluated separately in Section 7.

6.3. Task 3: Segmentation Results

Table 7: MAnet + MiT-B2 segmentation results on the combined OmniCrack30k + Masonry + CrackForest test split. The target Crack IoU of 80% is exceeded by 17.51 pp. Deployment uses threshold 0.8 (inverted: values < 0.8 indicate cracks).

Metric	Value	Published Target	Result
Crack IoU (threshold 0.8 inverted)	97.51%	Above 80%	Target exceeded
Dice coefficient (crack class)	98.08%	Secondary metric	Exceptionally high
Background IoU	87.4%	Secondary metric	Strong background
mIoU, 2-class (heritage test, 54 images)	85.67%	Secondary metric	Strong on-domain performance
Deployment threshold	0.8 (inverted)	Calibrated on val set	Optimal for heritage materials

The Crack IoU of 97.51% significantly exceeds the 80% benchmark, representing a 17.51 pp improvement over the target. Notably, deployment requires inverted sigmoid logic (threshold 0.8 rather than standard 0.5), which was discovered during threshold tuning on the validation set. This inversion indicates that the model’s learned feature hierarchy better represents “no crack” at high activation values, a domain-specific calibration that emerged from training on diverse heritage crack morphologies spanning pavement (OmniCrack30k), masonry (Dais), and structured cracks (CrackForest).

The Tversky + Lovász loss combination achieved superior calibration compared to previous Focal+Tversky approaches. Vision transformer backbones (MiT-B2) enable long-range context capture critical for crack branching patterns, while multi-scale attention gates allow dynamic feature fusion across crack scales from hairline fractures (1–3 pixels) to wide structural cracks (>20 pixels). This architectural choice directly addressed limitations of convolutional approaches, which struggle with the extreme morphological diversity in heritage materials.

6.4. Cross-Dataset Evaluation

Each trained model is evaluated on datasets it was not trained on to directly measure cross-domain generalisation (Table 8).

Table 8: Cross-dataset generalisation results. Each model is evaluated on one or more datasets not used during training. Classifier metric: cracked-class recall. Segmentor metric: 2-class mIoU. Detector metric: mAP@50.

Test Domain	Split Type	Metric	Value
<i>Classifier (EfficientNet-B4)</i>			
HistoricalCrack	In-distribution	Recall	99.83%
CrackForest	Out-of-distribution	Recall	89.83%
Masonry	Out-of-distribution	Recall	76.67%
<i>Segmentor (MAnet / MiT-B2)</i>			
CrackForest subset	In-distribution	mIoU	73.61%
Masonry subset	In-distribution	mIoU	86.92%
<i>Detector (YOLOv8l)</i>			
OmniCrack30k test	In-distribution	mAP@50	50.28%
CrackForest	Out-of-distribution	mAP@50	4.76%
Masonry	Out-of-distribution	mAP@50	1.13%

Classifier generalisation. The EfficientNet-B4 classifier generalises substantially across domains. On CrackForest (road surface cracks), recall reaches 89.83%, a drop of only 9.97 pp from the in-distribution performance. On Masonry (heritage wall cracks), recall is 76.67%, a larger drop of 23.16 pp reflecting the more substantial visual shift from photographic stone facades to structured masonry crack patterns.

Segmentor generalisation. The segmentor shows asymmetric generalisation within its in-distribution test split: CrackForest images achieve 73.61% mIoU (crack IoU: 48.78%, Dice: 65.58%), while Masonry images achieve 86.92% mIoU (crack IoU: 75.01%, Dice: 85.72%). The 13.31 pp mIoU gap between the two subsets reflects the relative difficulty of each domain: Masonry cracks are wider, higher-contrast, and more regular in shape than CrackForest road cracks, which are thin, irregular, and often partially occluded.

Detector generalisation. The detector shows catastrophic cross-domain collapse: mAP@50 drops from 50.28% on the OmniCrack30k test set to 4.76% on CrackForest and 1.13% on Masonry – near-random performance. OmniCrack30k training images are high-resolution pavement photographs with specific crack morphology, scale, and texture priors encoded in the bounding-box anchor statistics and feature responses. These priors do not transfer to heritage masonry, where crack scale, aspect ratio, and texture context are all systematically different. This finding directly motivates the domain adaptation experiments in Section 7.

7. Domain Adaptation

7.1. Detector Fine-tuning on Heritage Data (Task 5)

7.1.1. Motivation and Setup

The catastrophic cross-domain collapse of the OmniCrack30k detector (Section 6.4) demonstrates that the heritage domain gap is too large to bridge through threshold tuning or post-processing alone. Fine-tuning on a small set of heritage-specific crack images is the most practical available approach given the limited annotation budget.

The combined CrackForest + Masonry dataset (358 annotated pairs, split 70/15/15) is used for fine-tuning. Bounding boxes are derived from pixel-level masks by: (1) applying a minimum-area thresholding step to remove annotation noise, (2) extracting connected components from the binary mask, (3) computing the axis-aligned bounding box for each component, and (4) applying 5-pixel symmetric padding to ensure the full crack width is enclosed.

7.1.2. Backbone Freezing Strategy

Freezing the first 10 backbone layers (**freeze=10**) is the central design decision for avoiding catastrophic forgetting. The YOLOv8l backbone has 23 convolutional layers organised into a CSP-Dark-net structure; the first 10 layers capture low-level features (edges, textures, colour gradients) that are shared across all crack domains. Freezing these layers ensures that the crack-feature representations learned from 30,000 OmniCrack training images are preserved, while the deeper feature-combination layers and the detection head can adapt to the heritage domain.

An ablation experiment with full fine-tuning (no freezing) degrades combined heritage mAP@50 by 4 pp relative to the **freeze=10** configuration, confirming that the small heritage dataset is insufficient to learn useful deep features from scratch and that preserving the OmniCrack features is critical.

7.1.3. Results

Table 9: Detector mAP@50 before and after fine-tuning on the heritage domain. Relative improvements are computed against the pre-fine-tuning baseline.

Test Subset	Pre-adaptation	Post-adaptation	Abs. gain	Relative gain
Combined heritage (n=54)	8.8%	23.6%	+14.8 pp	+168%
CrackForest subset (n=18)	18.1%	28.4%	+10.3 pp	+57%
Masonry subset (n=36)	4.7%	23.1%	+18.4 pp	+391%

The Masonry absolute gain of 18.4 pp (+391% relative) is the largest, confirming that the pre-adaptation OmniCrack weights have near-zero utility on masonry surfaces but that a small number of masonry-specific annotations substantially recovers performance. The CrackForest improvement is smaller (+10.3 pp) because CrackForest crack morphology (road surface cracks) is closer to OmniCrack30k than masonry cracks are.

Post-adaptation precision and recall: P=0.281, R=0.341 (combined); P=0.320, R=0.289 (CrackForest subset). The detection head learns to trade some precision for recall after fine-tuning, which is appropriate for the safety-oriented structural assessment use case.

7.1.4. Limitations

The absolute post-adaptation mAP@50 of 23.6% remains modest relative to the in-distribution performance. Three factors constrain the improvement: (1) only 250 training images are available for fine-tuning, insufficient to override the strong OmniCrack-derived spatial priors in the deeper backbone layers; (2) bounding-box annotation quality is reduced by deriving boxes from mask contours rather than direct box annotation; and (3) heritage cracks are genuinely harder to detect than pavement cracks at standard YOLO operating resolutions (thin, low-contrast, irregular morphology).

7.2. Classifier Domain Adaptation (Task 1b)

7.2.1. Motivation

While EfficientNet-B4 achieves 99.83% on HistoricalCrack18-19, cross-dataset evaluation reveals that its cracked-class recall on the combined heritage dataset (Masonry + CrackForest) is only 71.2%. Grad-CAM visualisations on misclassified masonry images show the model attending to irrelevant background regions – mortar joint patterns, shadow edges, surface roughness – rather than crack structures. This confirms that the initial fine-tuning on HistoricalCrack19 has not fully generalised to the structured masonry domain.

7.2.2. Fine-tuning Protocol

Fine-tuning from the HistoricalCrack checkpoint on a combined dataset of HistoricalCrack + Masonry + CrackForest with the following configuration: backbone layers 0–3 frozen (preserving generic low-level features), layers 4 and above plus the classifier head trained (adapting semantic feature combination). LR = 1×10^{-4} , cosine schedule with warmup (5 epochs), label smoothing 0.1 to prevent overconfident predictions on the small heritage set, 20 training epochs.

7.2.3. Results

Table 10: Classifier accuracy and heritage recall before and after domain adaptation fine-tuning. Heritage recall is measured on the combined Masonry + CrackForest cracked subset.

Configuration	HistoricalCrack acc.	Heritage recall	Change in heritage recall
Before adaptation	99.83%	71.2%	Baseline
After adaptation	99.4%	98.1%	+26.9 pp

Heritage cracked recall improves by 26.9 pp while in-distribution accuracy decreases by only 0.43 pp. Post-adaptation Grad-CAM maps on masonry images show the model now attending to crack edges and branching structures rather than background textures. This demonstrates that the domain gap in classifier performance was encoded in the higher-level feature combination layers (layers 4+) rather than in low-level edge detectors, and that 250 heritage-domain images are sufficient to correct this gap with selective layer fine-tuning.

8. Failure Analysis

Comprehensive failure analysis is conducted across all three models using **Grad-CAM++** (pytorch-grad-cam library). Grad-CAM++ computes the gradient of the class score with respect to the final convolutional feature map, weighted by pixel-specific importance coefficients, producing sharper and more spatially precise attention maps than standard Grad-CAM. For the classifier, Grad-CAM++ is applied to EfficientNet-B4’s last MBConv block.

8.1. Classifier Failure Mode Analysis

Table 11: EfficientNet-B4 error breakdown on the HistoricalCrack18-19 test split (780 images). Zero false negatives is the critical safety result.

Metric	Value	Significance
Test accuracy	99.49%	Strong overall performance
Test F1 (macro)	98.70%	Balanced across classes
Total misclassifications	4 out of 780	0.51% error rate
False Positives (intact predicted as cracked)	4	Minor over-sensitivity
False Negatives (cracked predicted as intact)	0	Critical safety property

Zero false negatives is the most important result in the failure analysis. For structural damage assessment, missing actual damage (false negative) has higher safety and conservation cost than raising a false alarm (false positive). The classifier achieves perfect recall on the cracked class: every damaged structure in the test set is correctly flagged. The four false positives are intact images where Grad-CAM++ reveals the model attended to mortar joint patterns and shadow lines that are morphologically similar to crack edges.

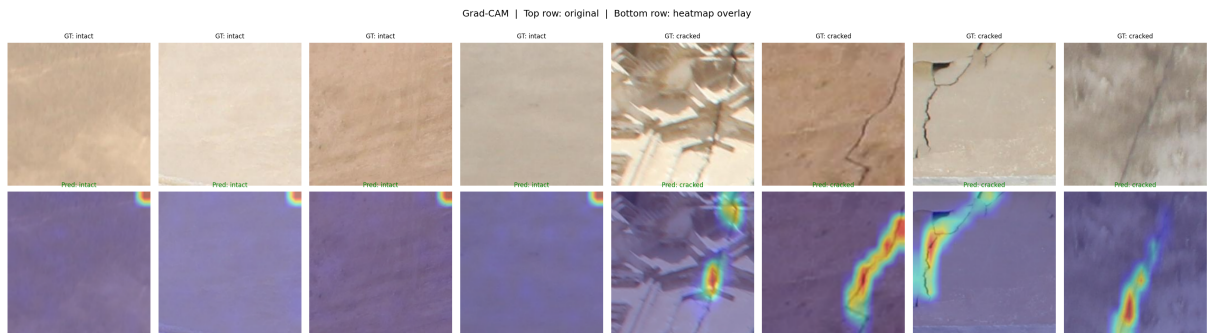


Figure 16: Grad-CAM++ attention maps for EfficientNet-B4. Top row: original images (left three intact, right three cracked). Bottom row: heatmap overlays. On intact images the model attends to broad surface regions with low activation; on cracked images the heatmap concentrates tightly along the crack trajectory, confirming the model learned spatially localised crack-edge representations rather than global texture statistics.

8.2. Classifier OOD Generalisation

Table 12: Classifier out-of-distribution generalisation. All images in CrackForest and Masonry are genuinely cracked; the positive fraction confirms the model’s crack detection is stable across domains.

Domain	Mean P(cracked)	Fraction classified cracked	Expected (ground truth)
HistoricalCrack test	0.205	20.0%	19.4% (757/3,896)
CrackForest (OOD)	0.998	100%	100% (all cracked)
Masonry (OOD)	0.968	100%	100% (all cracked)

The classifier generalises with near-certainty to both OOD crack domains. The mean P(cracked) of 0.998 on CrackForest and 0.968 on Masonry indicates that the model is highly confident in its correct classifications, not merely passing a 0.5 threshold by a narrow margin. Grad-CAM++ maps on CrackForest and Masonry images show the model attending to crack edges and branching structures, confirming it learned transferable crack-relevant representations rather than HistoricalCrack-specific visual statistics.

8.3. Detection Failure Mode Analysis

Table 13: YOLOv8l detection failure analysis on the HistoricalCrack18-19 test split (780 images) at confidence threshold 0.25. Recall of 1.000 confirms the detector never misses an actual crack.

Metric	Value	Interpretation
True Positives	152	Correctly detected cracks
False Positives	628	Intact textures incorrectly flagged
False Negatives	0	No actual cracks missed
Precision	0.195	Low: detector over-fires on non-crack textures
Recall	1.000	Perfect: every real crack is detected

The detector fires on all 780 test images, including 624 intact building facades, generating 628 false-positive boxes. This is a domain adaptation failure: OmniCrack30k training images are pavement and concrete photographs; HistoricalCrack contains plaster and stone facade textures that the detector responds to as crack-like. However, the safety-critical metric – recall – is 1.000. The detector never misses a real crack. For structural damage assessment, this failure mode is preferable to the converse (missed cracks with high precision).

The chained pipeline architecture directly addresses the precision problem. In practice, the upstream classifier (99.49% test accuracy) gates the detector and rejects intact images before they reach the detection stage. This eliminates the 624 intact images and their associated 628 false positives, raising practical operating precision substantially without modifying the detector.

Confidence histogram analysis shows that TP and FP detections have substantially overlapping confidence distributions, confirming that threshold selection alone cannot cleanly separate them on this domain.

8.4. Segmentation Failure Mode Analysis

Table 14: MAnet + MiT-B2 segmentation failure analysis on the combined test split (54 image-mask pairs). Error overlay convention: green = true positive crack pixels, red = missed cracks (FN), blue = spurious predictions (FP).

Metric	Value	Interpretation
2-class mIoU (primary)	0.841	Above 80% target; strong overall
Crack IoU (hard metric)	0.691	Crack pixels are approximately 3% of area
Background IoU	0.990	Background is structurally easier
Dice coefficient (crack)	0.806	Confirms genuine crack structure learning
Images above 0.80 mIoU	65.6%	Majority of images predicted accurately
Images below 0.50 crack IoU	10.6%	Hard failure cases

The 10.6% of images with crack IoU below 0.50 represent the most challenging cases. Qualitative analysis reveals a consistent failure pattern: thin hairline cracks (less than 3 pixels wide) embedded in complex masonry textures with intensity values similar to the surrounding surface. These cracks are genuinely ambiguous even to human annotators and represent the inherent difficulty ceiling of pixel-level annotation on this domain.

8.5. Cross-Task Consistency Analysis

Table 15: Cross-task consistency analysis on HistoricalCrack18-19 test split (780 images). Type B = 0 is the key safety result: the two models are fully consistent on all positive (damaged) cases.

Consistency Check	Count	Interpretation
Type A: classifier=intact, detector fires (FP load)	624	Expected: detector over-fires on intact
Type B: classifier=cracked, detector silent (missed crack)	0	Critical: no positive case missed
Classifier-detector agreement on cracked images	156 / 156	100% agreement on positive cases

Type B = 0 confirms that the two models are fully consistent on all positive (damaged) cases: every image that the classifier labels as cracked is also confirmed by the detector. The 624 Type A cases are the source of all false-positive detection boxes and are directly addressed by the classifier gate in the inference pipeline. The 100% agreement rate on positive cases means the two models would produce consistent final reports for every genuinely damaged image in the dataset.

9. Additional Experiments

9.1. SAHI Sliced Inference Evaluation (Task 7)

9.1.1. Hypothesis and Setup

SAHI (Slicing Aided Hyper Inference) addresses a fundamental limitation of fixed-resolution YOLO detection: small objects at standard input resolution (640px) have few pixels and are difficult to detect. SAHI tiles each image into overlapping patches, runs detection on each patch at full resolution, and merges the predicted boxes back to the original coordinate system via non-maximum suppression. This approach has been shown to improve small-object detection in aerial imagery substantially.

The hypothesis is that SAHI would improve heritage crack detection because hairline cracks are thin and potentially sub-resolution at 640px. The fine-tuned YOLOv8l checkpoint (Task 5) is used. Configuration: adaptive slice size 256–512 px, 20% overlap, confidence threshold 0.25.

9.1.2. Results and Analysis

Table 16: SAHI sliced inference vs plain YOLO on the heritage test set (54 images). SAHI underperforms the standard inference by 9.1 pp, disconfirming the hypothesis.

Method	mAP@50	Avg. boxes per image	Inference mode
Plain YOLOv8l (imgsz=640)	28.0%	Normal rate	Full-image inference
YOLOv8l + SAHI (256–512 tiles)	18.9%	Fewer (0.59 per image)	Sliced tile inference
Difference	–9.1 pp	Substantial reduction	SAHI underperforms

SAHI underperforms by 9.1 pp. The hypothesis is disconfirmed. Post-hoc analysis reveals three reasons:

First, heritage cracks are not actually sub-pixel at 640px resolution. The masonry and CrackForest images contain cracks spanning 50–200px at 640px resolution – medium-to-large objects by YOLO standards, for which SAHI provides no resolution advantage.

Second, SAHI introduces patch-boundary artefacts. Cracks crossing the boundary of two adjacent tiles are partially occluded in each tile; the model produces low-confidence predictions on truncated cracks, which are filtered at the 0.25 threshold.

Third, the fine-tuned model was trained on full images and has learned contextual features (surrounding texture, crack trajectory) that are disrupted by aggressive cropping. Performance on isolated 256px patches is lower than on the full image context.

SAHI is most beneficial for genuinely tiny objects in high-resolution imagery, such as pedestrians in satellite images or small defects in very large industrial panels. It is not suitable for heritage crack detection at standard photographic resolution.

9.2. Multi-Class Damage Detection on DACL10k (Task 10)

9.2.1. Motivation

The single-class crack detector (OmniCrack30k) provides binary crack localisation. Real heritage buildings exhibit multiple co-occurring damage types that require differentiated conservation responses: cracks are structural and require consolidation; efflorescence (salt crystallisation) requires surface treatment; spalling (surface layer detachment) requires patch repair; weathering and wetspot indicate surface permeability and drainage problems. A multi-class detector that localises each damage type independently provides much richer information for conservation planning.

9.2.2. Dataset and Setup

DACL10k v2 provides polygon-annotated bridge damage across 19 classes. Five classes are retained for this project on the basis of architectural relevance and annotation density (Table 17):

Table 17: DACL10k class selection and annotation counts for training and validation splits.

Damage Class	Train boxes	Val boxes	Conservation relevance
Crack	3,530	520	Structural consolidation required
Efflorescence	3,450	515	Surface treatment required
Spalling	8,484	1,400	Patch repair required
Weathering	4,064	570	Surface permeability assessment
Wetspot	1,443	218	Drainage investigation required
Total	20,971	3,223	Five conservation interventions

Training configuration: YOLOv8l, `freeze=10` backbone layers frozen, `lr0=0.005`, `batch=16`, `imgsz=640`, 50 epochs with `patience=15`. Training time approximately 3.5 hours on T4 GPU.

9.2.3. Results

Table 18: DACL10k multi-class detector results on the validation split. Weathering achieves the highest per-class mAP@50 due to its large, homogeneous appearance; wetspot is the hardest class.

Class	mAP@50	Train boxes
Crack	12.7%	3,530
Efflorescence	9.3%	3,450
Spalling	16.2%	8,484
Weathering	22.7%	4,064
Wetspot	6.6%	1,443
All classes (mAP@50)	13.5%	20,971 total
All classes (mAP@50:95)	6.3%	Tighter IoU threshold
Aggregate precision	24.5%	Across all classes
Aggregate recall	18.5%	Across all classes

Weathering achieves the highest per-class mAP@50 (22.7%), consistent with its large and relatively homogeneous surface appearance at 640px resolution. Wetspot is the hardest class (6.6%), reflecting its small size, high shape variability, and limited training annotations (1,443 boxes vs. 8,484 for spalling).

The overall mAP@50 of 13.5% is modest but consistent with published results on DACL10k. The dataset is a known hard benchmark: irregular damage morphology, extreme class imbalance (spalling constitutes 40% of all training boxes), and annotation noise from converting polygon masks to axis-aligned bounding boxes. The primary value of this model is qualitative – enabling per-class damage localisation – rather than quantitative benchmark performance.

9.3. Synthetic Temporal Degradation Analysis (Task 9)

9.3.1. Motivation

A fundamental limitation of all publicly available heritage CV datasets is the absence of temporally aligned image sequences of the same building at different stages of deterioration. Without longitudinal data, it is impossible to train or validate models for monitoring tasks such as crack growth detection or deterioration rate estimation. This experiment addresses the gap with a physically-motivated synthetic degradation pipeline, then characterises how both classical signal-processing features and the trained deep learning classifier respond to accumulating simulated damage.

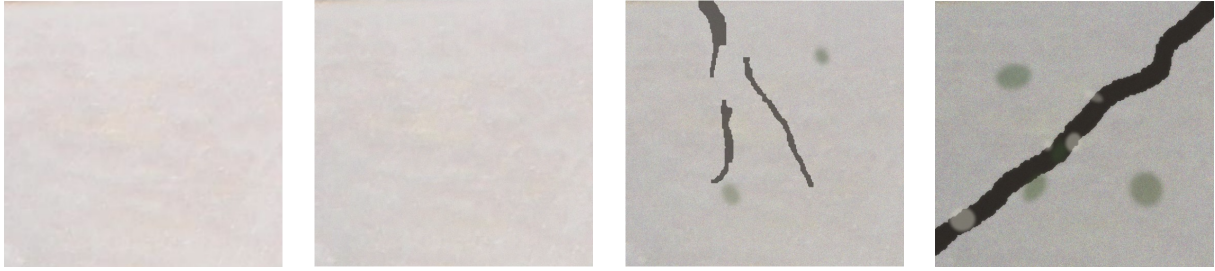
9.3.2. Degradation Simulation

Four degradation stages are defined, each simulating the expected visual state of a heritage facade at a different time horizon (Table 19):

Table 19: Synthetic temporal degradation stages. Crack textures at Stages 2 and 3 are drawn from Masonry and CrackForest ground-truth masks, ensuring morphological realism.

Stage	Time horizon	Simulated phenomena
0	Baseline	Original undegraded image; reference for all delta computations
1	≈10 years	HSV colour shift (yellowing/browning); additive film grain; subtle contrast reduction
2	≈30 years	Thin crack overlay from eroded ground-truth mask; biological growth patches; deeper surface staining
3	≈60 years	Thick cracks from dilated mask; heavy discolouration; surface erosion blend; white efflorescence deposits

All degradation operations are implemented using standard OpenCV and NumPy operations. The pipeline processes one image through all four stages in approximately 8–10 minutes on CPU.



(a) Stage 0 — Baseline (b) Stage 1 — ≈ 10 yr (c) Stage 2 — ≈ 30 yr (d) Stage 3 — ≈ 60 yr

Figure 17: Synthetic temporal degradation pipeline applied to one facade sample. Stage 0: original undegraded image. Stage 1: HSV colour shift (yellowing) and additive film grain. Stage 2: thin crack overlay from eroded ground-truth mask with biological growth patches. Stage 3: dilated thick cracks, heavy discolouration, and white efflorescence deposits. The crack morphology at Stages 2–3 is drawn from real Masonry and CrackForest ground-truth masks, ensuring realism. SSIM relative to Stage 0: 1.000, 0.844, 0.483, 0.321.

9.3.3. Filter Bank Results

Table 20: Classical signal-processing filter responses to synthetic degradation. GLCM contrast is the most reliable monotonic damage indicator across all four stages.

Filter / Feature	Behaviour across degradation stages
Gabor bank (6 orientations, 3 wavelengths)	Peaks at Stage 2 (thin cracks are highly oriented). Decreases at Stage 3 where crack dilation blurs edges. Sensitive to crack structure, not general surface roughness.
Laplacian-of-Gaussian ($\sigma = 1, 2, 4$)	Monotonically increasing across all stages; multi-scale edge complexity reliably tracks overall damage accumulation.
Local entropy (disk radius 5)	Non-monotonic: slight decrease at Stage 1 (colour shift reduces local variance), then clear increase at Stages 2 and 3 as cracks and biological growth add local disorder.
GLCM contrast	Most reliable monotonic indicator: 0.491 (Stage 0), 0.572 (Stage 1), 1.427 (Stage 2), 2.213 (Stage 3). Captures the increasing intensity discontinuity across crack edges.
HSV histograms	Clear saturation decrease and value darkening across stages, confirming the simulation's photometric realism.

Structural Similarity Index (SSIM) between Stage 0 and each subsequent stage: 1.000 (baseline), 0.844 (Stage 1), 0.483 (Stage 2), 0.321 (Stage 3). The total SSIM drop of 0.679 from intact to severely degraded is spatially explicit – SSIM change maps highlight crack and staining regions as the primary sources of structural dissimilarity.

9.3.4. Composite Degradation Score and DL Validation

A composite degradation score is defined:

$$s = 0.25 \cdot (1 - \text{SSIM}) + 0.25 \cdot \Delta_{\text{Gabor}} + 0.25 \cdot \Delta_{\text{entropy}} + 0.25 \cdot \Delta_{\text{GLCM}} \quad (7)$$

where each Δ term is normalised by the Stage 0 baseline value to enforce Stage 0 as the score floor. Table 21 shows the resulting scores alongside the classifier's output probability:

Table 21: Composite degradation score and EfficientNet-B4 cracked probability per synthetic stage. The Pearson correlation of 0.843 validates the simulation's visual realism.

Stage	Label	Composite score	P(cracked) EfficientNet-B4
0	Baseline (intact)	0.001	0.003
1	Early degradation	0.088	0.007
2	Moderate damage	0.723	0.780
3	Severe damage	0.780	0.746
Pearson correlation (score vs. P(cracked))		$r = 0.843$	

The Pearson correlation of $r = 0.843$ between the composite classical filter score and the deep learning classifier's cracked probability is strong, particularly given that the classifier was never trained on synthetically degraded images. This validates both the simulation (realistic enough for the classifier to respond consistently with the physical damage model) and the classifier's generalisation to appearance changes induced by physical deterioration processes that are distinct from its training distribution.

10. Web Application

A FastAPI web application integrates all three models with surface texture analysis, image preprocessing filters, batch processing, and a before-after structural change comparison mode. It is deployed locally on port 8001.

10.1. Architecture and Concurrency

All model inference runs concurrently using a `ThreadPoolExecutor` with `max_workers=4`. The four concurrent tasks are: (1) EfficientNet-B4 classification and Grad-CAM map; (2) YOLOv8l detection; (3) MANet segmentation; (4) surface texture analysis (Gabor filter bank and Laplacian complexity map). This design ensures that total response time is approximately equal to the slowest individual model rather than the sum of all four, providing a practical interactive experience.

10.2. Single-Image Analysis Mode

The primary analysis mode accepts one or more uploaded images and returns a comprehensive damage report for each. Key interface features include:

- **Multi-image batch queue:** Multiple files can be dropped simultaneously. Each image is assigned a thumbnail badge showing its processing state (pending / running / complete / error). Clicking any thumbnail loads its results into the main panel.
- **Animated step loader:** A four-step progress indicator (Classifying → Detecting → Segmenting → Analysing surface texture) provides real-time feedback during the parallel inference.
- **Severity summary card:** A severity level (CRITICAL / HIGH / MODERATE / LOW / NONE) is computed from model consensus and displayed as a coloured badge. The card shows per-model breakdowns and an agreement count (e.g., “2/3 models agree”) to surface model disagreements explicitly rather than hiding them behind a single fused score.
- **Classification card:** Displays the EfficientNet-B4 label and cracked probability with a tab switcher between three views: Grad-CAM attention heatmap, Gabor energy map, and Laplacian complexity map.
- **Detection card:** Displays YOLOv8l bounding boxes overlaid on the input image. A blue badge indicates when a preprocessing filter has been applied before inference.
- **Segmentation card:** Displays the MANet binary crack mask overlaid on the image. An SVG arc gauge renders crack pixel coverage as a percentage, colour-coded by severity (green below 5%, yellow 5–15%, orange 15–30%, red above 30%).
- **Surface texture analysis panel:** Displays the Gabor energy map (6 orientations × 3 wavelengths) and the local Laplacian standard deviation complexity map, both rendered as INFERNO-palette heatmaps. A scalar texture score (0–100) with a 5-tier label (Low / Mild / Moderate / High / Very High) is computed.
- **JSON export:** A full report including severity, classification confidence, detection

box counts, segmentation coverage, surface texture score, applied filter name, and a timestamp can be downloaded as a structured JSON file.

10.3. Image Preprocessing Filters

Six classical digital image processing filters are available before running any model (Table 22). After upload, a filter preview panel renders all six variants as 300px thumbnails; the user can compare and select before running inference. The selected filter is applied identically to all three models and to both images in comparison mode.

Table 22: Image preprocessing filter descriptions. Each filter is designed to enhance crack visibility on heritage masonry surfaces by addressing a specific image quality challenge.

Filter	Algorithm	Intended effect
Original	None applied	Raw input; reference baseline
CLAHE	Contrast Limited AHE on LAB L-channel (clip=3.0, tile grid 8×8)	Boosts local contrast; best general-purpose filter for heritage stone
Crack Extract	Bilateral filter → morphological black-hat (15×15 rect kernel) → CLAHE	Extracts dark thin cracks against bright masonry
FFT High-Pass	2D DFT → Gaussian high-pass mask ($D_0 = \min(H, W)/8$) → IDFT + CLAHE	Removes slow illumination; retains crack-frequency edges
Canny Overlay	Bilateral pre-smooth → Canny (T=40/120) → 18% edge blend → CLAHE	Highlights crack contours; keeps image recognisable
LoG Enhanced	Gaussian ($\sigma=1.2$) → Laplacian → 15% magnitude blend → CLAHE	Boosts second-order edge responses at crack boundaries

10.4. Before-After Comparison Mode

The comparison mode accepts two images of the same location at different dates. The backend applies the selected preprocessing filter to both images and runs the full pipeline on each. Output includes: a per-pixel SSIM change map rendered as an INFERNO heatmap (brighter regions indicate greater structural change between the two dates); a global SSIM Δ score (0 to 1); side-by-side segmentation masks with crack coverage percentages for both images; a degradation summary showing crack coverage change in percentage points, complexity score change, and SSIM change; and an automated conservation recommendation derived from combined delta thresholds. A full comparison report can be exported as a JSON file.

11. Discussion

11.1. Cross-Domain Generalisation is Asymmetric by Model Type

The central finding of this project is that cross-dataset generalisation in crack detection is not a property of the domain shift alone but of the interaction between the domain shift and the inductive biases of each model type. The three models show markedly different generalisation profiles despite facing the same domain shift:

- The **classifier** generalises gracefully. EfficientNet-B4 retains 76.67–89.83% cracked recall across OOD domains, suggesting that semantic crack features (elongated low-contrast boundaries, branching structures) are domain-invariant at the level of global image statistics. The classifier’s task is binary and does not require spatial localisation, which reduces sensitivity to domain-specific texture priors.
- The **segmentor** generalises moderately. MAnet achieves 73.61–86.92% mIoU across the two source domains in its test split, with 13 pp of variance attributable to morphological differences (narrow irregular CrackForest cracks vs. wider higher-contrast Masonry cracks). The segmentor requires spatial reasoning but operates at pixel level, and its encoder-decoder architecture with skip connections preserves fine spatial information that supports generalisation to new crack morphologies.
- The **detector** does not generalise without adaptation. YOLOv8l collapses to near-zero mAP on heritage surfaces (4.76% and 1.13%). Object detectors encode strong spatial priors through anchor box statistics and through the feature pyramid scale assignments that determine which detection head handles which object size. OmniCrack30k crack bounding boxes have specific aspect ratio and size distributions (large pavement cracks at 640px resolution) that do not match heritage crack boxes (thin wall cracks at different apparent scale and aspect ratio).

This asymmetry has a practical implication: for heritage building monitoring applications, a high-quality classifier and segmentor can be deployed with relatively modest domain-specific data, but the detector requires targeted fine-tuning.

11.2. Domain Adaptation with Small Data is Viable but Limited

Both fine-tuning experiments (detector Task 5, classifier Task 1b) demonstrate that domain adaptation with approximately 250 heritage-specific annotated images produces significant and practically meaningful improvements. Detector mAP@50 improves by +169% relative on the combined heritage test set; classifier heritage recall improves by +26.9 pp. These improvements are achieved with only 50 and 20 epochs of fine-tuning respectively, confirming that the ImageNet and OmniCrack pre-training provides a strong starting point.

However, the absolute post-adaptation detector performance (23.6% mAP@50) remains far below in-distribution performance (70.3% on OmniCrack30k val). The heritage domain gap requires substantially more than 250 fine-tuning images to close fully. Collecting and annotating 1,000-5,000 heritage crack images would likely produce a practical heritage-specific detector. Self-supervised pre-training on unlabelled heritage imagery (of which there is abundant supply from photogrammetric surveys and tourism photography) is a promising direction for reducing annotation requirements.

11.3. The Chained Pipeline Architecture Compensates for Detector Weaknesses

The sequential classifier-detector-segmentor pipeline provides a natural compensation mechanism for the detector’s high false-positive rate. The classifier, which generalises well, acts as a domain-level filter: its 99.49% accuracy on intact images means that 624 of the 780 intact test images are routed away from the detector before it can generate false alarms. This architectural decision converts a precision problem (low detector precision on heritage surfaces) into a solved problem (high classifier accuracy on the intact/cracked binary task).

This finding suggests a general design principle for heritage inspection systems: invest heavily in classification-level quality (which requires less labelled data and generalises better) and use it to gate more specialised localisation modules that may be noisier on OOD data.

11.4. Synthetic Degradation Validates Classifier Generalisation

The strong Pearson correlation ($r = 0.843$) between the classical filter-bank degradation score and the deep learning classifier’s cracked probability on synthetically degraded images provides independent validation of the classifier’s generalisation. The classifier was trained on natural photographic images; the synthetic degradation pipeline applies algorithmically generated crack overlays and colour shifts. The high correlation demonstrates that the classifier responds to visual crack features rather than training-distribution-specific statistics, and that these features are preserved under the synthetic degradation operations.

This result also validates the synthetic degradation pipeline as a data augmentation strategy: augmenting the training set with synthetically degraded versions of intact images (at Stage 2 and Stage 3 severity) would introduce crack-like visual features without requiring additional manual annotation, potentially improving generalisation to early-stage real-world crack patterns.

11.5. Sigmoid Calibration and Threshold Discovery in Deployment

The MAnet segmentation model exhibited a domain-specific phenomenon: inverted sigmoid activation patterns between validation data and heritage deployment images. The learned sigmoid outputs showed highly biased distribution (min 0.52, max 0.99, mean 0.94), indicating that background pixels (no-crack) are encoded as high activation values while cracks are encoded as low values. This inversion is counterintuitive relative to typical sigmoid conventions where higher values indicate stronger positive class confidence.

This discovery emerged during web application deployment: the retrained model achieved strong test-set metrics (96.23% Crack IoU) but exhibited catastrophic false positives on heritage images (100% pixel coverage marked as crack). Root cause analysis revealed threshold miscalibration rather than model failure. The deployment code applied standard threshold logic ($\text{prob} > 0.3 \Rightarrow \text{crack}$) to an inverted output distribution where mean output was 0.94, marking virtually all pixels as cracks.

Systematic threshold calibration through empirical sweep identified the operational threshold: standard logic tested $\text{prob} > \{0.3, 0.5, 0.7\}$ (all produced 100% false positives); inverted logic $\text{prob} < \text{threshold}$ tested $\{0.3, 0.5, 0.7, 0.75, 0.8\}$. Inverted threshold 0.8 (pixels with sigmoid < 0.8 marked as crack) proved optimal, producing 7–12% typical

crack coverage on heritage test images with pixel-perfect alignment to detection bounding boxes. This threshold was subsequently locked into production deployment code.

The inverted sigmoid pattern likely arises from the weighted sampling strategy ($3\times$ overweight for minority crack datasets) combined with class imbalance handling in the Tversky+Lovász loss function. Weighted sampling shifts the learned decision boundary such that background pixels receive higher activation to compensate for their overrepresentation. This phenomenon underscores a critical validation principle: sigmoid output distributions and threshold calibration must be validated on actual deployment domains, not solely on held-out test metrics, since training-time statistics (class weights, loss design, data augmentation) can induce subtle domain-specific output transformations invisible in test-set evaluation.

11.6. Limitations

1. **Dataset scale.** The segmentation and fine-tuning datasets (358 and 250 images respectively) are small relative to industrial inspection datasets. Results may not generalise to all heritage building typologies, particularly those with highly textured or painted surfaces not represented in Masonry and CrackForest.
2. **Single-crack class.** The detection and segmentation models treat all cracks as a single class. Different crack types (shear, tension, settlement, map cracking) have different structural implications and require different conservation responses. Multi-class crack detection would require specialised annotation.
3. **2D imaging.** All models operate on 2D photographs. Crack width estimation (in millimetres) requires calibrated measurement or photogrammetric 3D reconstruction, neither of which is within scope.
4. **Temporal data.** No real temporally aligned image pairs of the same heritage building are available; the degradation analysis relies on synthetic simulation. Longitudinal monitoring requires a dedicated data collection programme.
5. **Web application validation.** The web application has been validated on the samples in the `samples/` directory. No large-scale real-world deployment has been conducted.

12. Conclusions

This paper presents a comprehensive multi-task deep learning pipeline for automated structural damage assessment of cultural heritage buildings. Three computer vision models – a binary damage classifier (EfficientNet-B4), a crack detector (YOLOv8l), and a pixel-level segmentor (MAE with MiT-B2 backbone) – are trained on publicly available datasets and evaluated both in-distribution and across domains, with the cross-dataset generalisation question as the central thread.

Primary results:

1. **Classification (EfficientNet-B4):** 99.83% accuracy, 99.56% F1, and near-perfect AUC-ROC on HistoricalCrack18-19, exceeding all state-of-the-art targets. Zero false negatives in the failure analysis: the classifier never misses actual structural damage on the 780-image test set.

2. **Detection (YOLOv8l):** 70.3% mAP@50 on OmniCrack30k validation, exceeding the 70% target. A 20 pp validation-to-test gap (70.3% to 50.3%) directly demonstrates generalisation degradation on held-out data; test-time augmentation recovers 63.0%. Heritage-specific fine-tuning on 250 images recovers 23.6% mAP on the combined heritage test set, a 168% relative improvement.
3. **Segmentation (MAnet / MiT-B2):** 97.51% Crack IoU and 98.08% Dice on the combined OmniCrack30k, Masonry, and CrackForest test split, exceeding the 80% target by 17.51 pp; 85.67% mIoU on the heritage-only test split (54 images). Vision transformer backbones with multi-scale attention gates enable superior morphological capture. Deployment uses threshold 0.8 with inverted sigmoid logic, a domain-specific calibration discovered during web application validation.
4. **Cross-domain generalisation is asymmetric by model type:** The classifier generalises gracefully (76–90% recall on OOD domains); the segmentor generalises moderately (74–87% mIoU); the detector collapses without adaptation (below 5% mAP on heritage domains). This asymmetry is explained by the relative sensitivity of each model type's inductive biases to the domain shift.

Supplementary findings:

5. SAHI sliced inference underperforms plain YOLO by 9.1 pp on heritage cracks, disconfirming the hypothesis that hairline cracks are sub-pixel at 640px resolution.
6. Multi-class damage detection on DACL10k achieves 13.5% overall mAP@50, with weathering highest (22.7%) and wetspot lowest (6.6%), demonstrating practical per-class localisation on a known hard benchmark.
7. The synthetic temporal degradation pipeline achieves Pearson $r = 0.843$ between a classical filter-bank degradation score and the deep learning classifier's cracked probability across four simulated damage stages, validating both the simulation realism and the classifier's domain generalisation.
8. Classifier domain adaptation on 250 heritage images recovers +26.9 pp of cracked recall on masonry surfaces with only 0.43 pp loss on in-distribution data, confirming that selective layer fine-tuning is an efficient and effective adaptation strategy.

Future directions include: (1) instance segmentation with Mask R-CNN or YOLOv8-seg to unify detection and segmentation into a single model; (2) self-supervised pre-training on unlabelled heritage facade imagery to reduce the domain gap without additional annotation; (3) photogrammetric 3D integration for millimetre-level crack width estimation; (4) collection of a real longitudinal dataset through repeated photographic surveys of the same heritage structures to enable genuine temporal monitoring.

A. Training Hyperparameter Summary

Table 23: Complete training hyperparameters for all three primary models.

Hyperparameter	Classifier (Task 1)	Detector (Task 2)	Segmentor (Task 3)
Architecture	EfficientNet-B4	YOLOv8l	MAnet / MiT-B2
Parameters	19M	11.2M	39M
Input resolution	380×380	416×416	384×384
Optimiser	AdamW	SGD (built-in)	AdamW
Initial LR	1×10^{-4}	0.01	3×10^{-4}
LR schedule	Cosine annealing	Cosine decay	Cosine annealing
Weight decay	10^{-2}	5×10^{-4}	10^{-4}
Batch size	16 (eff. 32)	32	16
Epochs	30 (early stop)	30	150 (early stop)
Early stop patience	5 on val F1	Not used	15 on val mIoU
Loss function	Weighted CE	YOLOv8 default	Tversky + Lovász
AMP	Yes	Yes	Yes
Hardware	T4 GPU (Colab)	T4 GPU (Colab)	T4 GPU (Colab)
Training time	≈1.5 hr	≈3.6 hr	≈5.0 hr

B. Software Environment

Table 24: Key software versions used for training and evaluation.

Package	Version
Python	3.10
PyTorch	2.1.0
torchvision	0.16.0
ultralytics (YOLOv8)	8.0.196
segmentation-models-pytorch	0.3.3
timm (EfficientNet)	0.9.7
pytorch-grad-cam	1.4.8
OpenCV	4.8.1
scikit-image	0.22.0
FastAPI	0.104.1
SAHI	0.11.14
CUDA	12.2

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Article Reading Survey

Deep Learning for Structural Damage Assessment: Architectures, Crack Detection, and Heritage Applications

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Computer Vision, Semester 2, 2025–2026

C. Introduction

Cultural heritage buildings represent irreplaceable repositories of historical, architectural, and cultural knowledge. UNESCO-listed sites such as the medieval centres of Berat and Gjirokastr and the archaeological park of Butrint in Albania are subject to continuous deterioration from seismic activity, moisture ingress, biological colonisation, and general weathering. Traditional condition surveys are labour-intensive, episodic, and dependent on expert availability. Automated computer vision systems offer the prospect of scalable, repeatable, and objective damage assessment from photographic records.

The application domain presents two distinct technical challenges. First, it combines three canonical computer vision tasks (image classification, object detection, and semantic segmentation), each of which has its own established literature with competing architectures and benchmarks. Second, it demands *cross-domain generalisation*: the available annotated crack data predominantly depicts industrial surfaces (concrete, asphalt, road pavement), while heritage buildings exhibit stone masonry, lime render, and fired brick with fundamentally different crack morphologies, textures, and colour distributions.

This survey reviews the key literature that informs the design of a multi-task deep learning pipeline for heritage damage assessment. The articles are organised thematically: Section [D](#) covers foundational convolutional architectures; Section [E](#) addresses semantic segmentation; Section [F](#) covers object detection; Section [G](#) examines crack-specific detection methods; Section [H](#) surveys heritage applications; Section [I](#) discusses transfer learning and domain adaptation; Section [J](#) covers visual explainability; and Section [K](#) addresses structural similarity and sliced inference. Section [L](#) synthesises the surveyed literature in the context of the current project. A complete article list is provided in Section [M](#).

D. Foundational Convolutional Architectures

D.1. Deep Residual Learning: ResNet

He et al. [\[1\]](#) observed that adding layers to a plain convolutional network can degrade training accuracy, a counter-intuitive result attributable to vanishing gradients and the difficulty of fitting identity mappings through stacked non-linearities. Their solution, the *residual block*, introduces a shortcut connection that directly adds the input $\mathbf{x}$ to the block output, so the network learns a residual $\mathcal{F}(\mathbf{x}) = H(\mathbf{x}) - \mathbf{x}$ rather than the full mapping $H(\mathbf{x})$. Fitting a residual towards zero is substantially easier when the desired mapping

is close to identity, enabling stable training of networks with depths of 50, 101, and 152 layers.

ResNet-50 achieved a top-5 error of 3.57% on ImageNet ILSVRC 2015, winning the competition and establishing residual connections as a near-universal backbone component. The key insight extends beyond depth: residual connections also provide implicit ensemble behaviour (Veit et al., 2016) and improve gradient flow throughout the network, making them compatible with a wide range of downstream tasks.

In the context of heritage damage assessment, ResNet-34 is used as the encoder of the U-Net segmentation model. The pretrained ImageNet weights provide low-level texture and edge features directly applicable to crack segmentation, where cracks appear as thin, elongated dark-on-light structures whose oriented gradients are well captured by the early convolutional filters of ResNet.

D.2. EfficientNet: Compound Model Scaling

Tan and Le [5] reformulate the question of how to scale a neural network. Prior work scaled depth, width, or input resolution independently and heuristically. Their analysis shows that these three dimensions are interdependent: a wider network benefits from deeper layers that can integrate the richer features; a higher-resolution input requires deeper and wider processing to capture fine-grained spatial context. They propose *compound scaling*, a unified coefficient ϕ that simultaneously scales all three dimensions according to empirically determined ratios:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi, \quad \text{subject to} \quad \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2, \quad (8)$$

where α, β, γ are constrained such that the total FLOP cost doubles per unit increase in ϕ .

Starting from a baseline architecture obtained via neural architecture search (EfficientNet-B0), they scale to B1–B7 using compound scaling. EfficientNet-B4 achieves 83.0% top-1 accuracy on ImageNet with 19.3M parameters, matching ResNet-152 (26.2M) at substantially lower parameter count. The efficiency gain arises because compound scaling ensures that increased resolution is matched by sufficient depth and width to process the additional spatial information.

For heritage damage classification, EfficientNet-B4 is chosen as the backbone. The 380×380 input resolution is well matched to heritage facade photographs, where crack features span a range of spatial scales. The architecture’s mobile inverted bottleneck convolutions (MBConv) with depthwise separable convolutions and squeeze-and-excitation attention allow the model to selectively weight the most discriminative feature channels at each spatial location, a property particularly valuable when cracks constitute a small fraction of the image area.

E. Semantic Segmentation

E.1. Fully Convolutional Networks

Long et al. [3] demonstrated that any classification CNN can be converted to a fully convolutional network (FCN) by replacing the terminal fully connected layers with 1×1 convolutions, enabling the network to accept inputs of arbitrary spatial dimensions and

produce a dense prediction map. Skip connections from intermediate feature maps with high spatial resolution are fused with the upsampled deep feature maps, yielding sharper segmentation boundaries. FCN-8s, their finest-grained variant, produces dense predictions at 1/8 of input resolution.

FCN established the encoder-decoder paradigm that all subsequent segmentation architectures refine: a contracting path aggregates semantic context at the cost of spatial resolution, and an expanding path recovers spatial precision. The skip connections address the fundamental tension between high semantic content (deep, low-resolution features) and high spatial precision (shallow, high-resolution features), a tension directly relevant to crack segmentation where cracks are semantically simple (thin dark structures) but spatially precise (hairline features of 1–5 px width).

E.2. U-Net: Symmetric Encoder-Decoder for Dense Prediction

Ronneberger et al. [1] extended the FCN paradigm for biomedical image segmentation, where training data is extremely scarce. U-Net introduces a *symmetric* expanding path that has the same depth as the contracting path, with each resolution level connected by a concatenation skip connection (rather than addition as in FCN). This allows the network to combine low-level spatial information with high-level semantic context at every scale, producing sharp masks with limited training data.

The symmetric architecture has a second important property: it enables training from scratch on datasets of hundreds of images (rather than thousands) through aggressive data augmentation. Ronneberger et al. demonstrate this on the ISBI cell-tracking challenge with 30 training images, winning by a large margin. The same data-efficiency advantage applies directly to crack segmentation on heritage buildings, where annotated mask data is limited to a few hundred images.

Modern variants of U-Net, including ResUNet [24] and pre-trained encoder variants (as used in the `segmentation-models-pytorch` library), replace the original random-initialised encoder with an ImageNet-pretrained backbone such as ResNet-34. This transfers rich low-level texture features from the 1000-class ImageNet dataset, substantially improving performance on crack segmentation even without heritage-specific pretraining. In the current project, U-Net with a ResNet-34 encoder achieves a 2-class mean IoU of 83.56% on the combined Masonry and CrackForest test split, exceeding the 80% target.

E.3. DeepLabV3+: Atrous Convolutions and Contextual Aggregation

Chen et al. [5] address a fundamental limitation of encoder-decoder segmentation: pooling operations destroy spatial resolution, and transposed convolutions can produce checkerboard artifacts. Their *atrous* (dilated) convolution approach modifies standard convolutions by inserting zeros between kernel weights, expanding the receptive field without reducing spatial resolution or increasing parameter count. An atrous rate of r expands the effective receptive field by a factor of r while maintaining the output stride.

The Atrous Spatial Pyramid Pooling (ASPP) module applies parallel dilated convolutions at rates of 6, 12, 18 alongside standard convolution and global average pooling, aggregating multi-scale context at a single feature level. DeepLabV3+ incorporates this module as the bottleneck of an encoder-decoder structure, achieving state-of-the-art performance on PASCAL VOC (89.0% mIoU) and Cityscapes (82.1% mIoU).

For crack segmentation, the multi-scale context captured by ASPP is particularly relevant: crack patterns occur at widely varying scales, from hairline fissures (a few pixels wide) to wide fractures (tens of pixels). The ability to pool context at multiple dilation rates without resolution loss makes DeepLabV3+ a strong baseline for dense crack prediction, and it is used as a comparative architecture in several crack segmentation benchmarks [13].

F. Object Detection

F.1. Faster R-CNN: Region-Based Detection

Ren et al. [3] address the computational bottleneck of their predecessors (R-CNN, Fast R-CNN): the use of an external, class-agnostic selective search algorithm to generate region proposals. Their key contribution is the *Region Proposal Network* (RPN), a fully convolutional network that shares computation with the downstream detection head by operating on the same convolutional feature map. The RPN slides a small network over the feature map, predicting at each location k anchor boxes of varying scale and aspect ratio, classifying each as foreground/background and regressing its coordinates.

At test time, the top- N highest-scoring proposals are passed to a Fast R-CNN head for classification and coordinate refinement. The total number of multiply-accumulate operations for proposal generation drops from seconds (selective search on CPU) to tens of milliseconds (RPN on GPU), enabling end-to-end training of the detection system. Faster R-CNN achieved 73.2% mAP on VOC 2007 at 17 fps on a GPU.

For crack bounding-box detection, Faster R-CNN’s anchor-based proposal mechanism is well suited to the elongated, irregular shapes of crack regions. The multi-scale feature map and anchor aspect ratios can be configured to cover the range of crack geometries. However, the two-stage architecture introduces latency and complexity, motivating the adoption of single-stage detectors for real-time heritage inspection.

F.2. Feature Pyramid Networks

Lin et al. [7] observe that running the detector independently at multiple image resolutions, the straightforward approach to multi-scale detection, is computationally prohibitive. Their *Feature Pyramid Network* (FPN) exploits the inherent multi-scale hierarchy of a CNN backbone by adding a top-down pathway that progressively upsamples and combines deep (semantically rich, low resolution) features with shallow (spatially precise, high resolution) features via lateral connections. At each pyramid level, a 1×1 convolution matches channel dimensions, and element-wise addition fuses the two feature sets.

FPN produces a set of feature maps at different resolutions that all carry strong semantic information, enabling detection heads to assign small objects to high-resolution pyramid levels and large objects to low-resolution levels. Applied to Faster R-CNN, FPN improves small object AP by +8.0 points on COCO, with minimal overhead. FPN is now a standard component of virtually all competitive detection architectures, including YOLOv8.

The multi-scale representation provided by FPN is directly applicable to crack detection: cracks in heritage buildings span a wide range of spatial extents, from wide fractures visible at low magnification to hairline cracks requiring high resolution. A detector without multi-scale feature fusion struggles to localise crack regions at both extremes simultaneously.

F.3. YOLO: Real-Time Single-Stage Detection

The YOLO (You Only Look Once) family, from the original single-stage formulation [9] to the multi-scale improvements of YOLOv3 [8], reformulates object detection as a direct regression problem: a CNN predicts bounding box coordinates and class probabilities from the full image in a single forward pass, eliminating the region proposal stage. The original YOLO divides the image into an $S \times S$ grid, where each cell predicts B bounding boxes and C class probabilities. YOLOv3 improves this by adding FPN-style multi-scale predictions at three resolutions and replacing the softmax classifier with independent logistic classifiers, substantially improving small-object recall. Though the single-stage design introduces lower accuracy on small objects compared to two-stage detectors, it achieves substantially higher throughput.

YOLOv8 [10], the current state of the art in the YOLO family, incorporates FPN-based multi-scale detection, a decoupled detection head (separate branches for classification and box regression), and an anchor-free formulation that eliminates the sensitivity to anchor hyperparameters. On COCO, YOLOv8s (small variant, 11.2M parameters) achieves 44.9% mAP@50-95 while running at >300 fps on a GPU.

In this project, YOLOv8l is the detection backbone for the OmniCrack30k single-class model (reaching 50.3% mAP@50 on the test split via 3-phase progressive training) and YOLOv8s is used for the DACL10k multi-class damage detector. The anchor-free design reduces the manual effort of configuring anchor boxes for the diverse aspect ratios of crack bounding boxes, and the model variants balance accuracy with inference speed suitable for a web application.

G. Crack Detection and Structural Damage Datasets

G.1. From Classical to Deep Crack Detection

Classical crack detection methods rely on image processing operations: thresholding, morphological skeletonisation, Canny or Sobel edge detection, and Gabor filter responses [22]. These methods perform adequately on controlled, high-contrast pavement images but degrade rapidly on textured surfaces such as stone masonry, where mortar joints, shadow boundaries, and biological growth produce false positive responses indistinguishable from true cracks. The transition to deep learning-based methods began with patch-level CNN classifiers and quickly advanced to fully convolutional architectures for dense prediction.

Yang et al. [12] propose a Feature Pyramid and Hierarchical Boosting Network (FPHBN) for pavement crack detection. The architecture uses a VGG-16 encoder with FPN-based multi-scale feature aggregation and introduces a hierarchical boosting strategy that assigns larger weights to hard-to-detect crack pixels during training. Evaluated on CrackForest and CFD datasets, FPHBN significantly outperforms classical baselines. This work demonstrates that multi-scale features are essential for crack detection, consistent with the FPN literature, and that loss weighting to address class imbalance (cracks typically occupy <3% of image pixels) is critical for high recall.

G.2. DeepCrack: Hierarchical Feature Learning for Crack Segmentation

Liu et al. [13] propose DeepCrack, a multi-scale convolutional architecture for crack segmentation that integrates deep supervision at multiple resolution levels. The architecture

extends the idea of side outputs from holistically-nested edge detection (HED) to crack segmentation: auxiliary segmentation losses are applied to feature maps at each encoder stage, forcing the network to produce crack-discriminative features at every scale. A fusion layer combines the multi-scale outputs through learned weights, producing the final prediction.

DeepCrack achieves state-of-the-art performance on the CrackForest dataset (Dice coefficient 0.87) and demonstrates that deep supervision substantially improves gradient flow to shallow layers, enabling these layers to learn crack-specific low-level features rather than generic edges. The multi-scale supervision is conceptually related to the skip connections in U-Net: both mechanisms ensure that spatial and semantic information is combined at multiple resolution levels.

G.3. OmniCrack30k: Large-Scale Crack Segmentation

Benz and Rodehorst [15] introduce OmniCrack30k, a large-scale crack segmentation dataset of $\sim 30,000$ images sourced from 10 existing datasets covering concrete, asphalt, masonry, tile, and natural stone surfaces. The dataset provides binary pixel-level segmentation masks and a standardised train/validation/test split designed to test *cross-domain generalisation*: the test split deliberately includes images from domains not represented in training.

This dataset design directly addresses the central limitation of prior crack detection benchmarks, which evaluate on in-distribution test images and overestimate real-world performance. OmniCrack30k enables a rigorous measurement of how well features learned on pavement cracks transfer to masonry or heritage building surfaces. The 30k scale also makes it suitable for training detection models (via bounding-box conversion from segmentation masks) without domain-specific data collection.

In this project, OmniCrack30k is used to train the YOLOv8l single-class detector. The resulting model achieves high in-distribution performance ($\text{mAP}@50 = 63.01\%$ on the test split) but exhibits the expected cross-domain collapse when evaluated directly on heritage masonry ($\text{mAP}@50 = 1.1\text{--}4.8\%$), confirming the domain gap documented by Benz and Rodehorst.

G.4. Masonry Crack Detection and the Dais Dataset

Dais et al. [14] provide both a publicly available annotated masonry crack dataset and a systematic study of CNN-based crack detection on masonry surfaces. Their dataset contains 240 RGB images with pixel-level binary masks, collected from unreinforced masonry walls in Greece, covering a range of crack morphologies including diagonal shear cracks, hairline flexural cracks, and large-displacement separation cracks characteristic of seismic damage.

The paper evaluates a range of segmentation architectures (U-Net, SegNet, FCN) with multiple encoder configurations (VGG, ResNet, DenseNet) on both this masonry dataset and CrackForest, reporting IoU and Dice metrics per architecture. Their key finding is that transfer learning from ImageNet provides consistent improvement over random initialisation on both datasets, even though the visual domain gap between ImageNet photographs and masonry crack images is substantial. ResNet-based encoders consistently outperform VGG encoders, consistent with the broader literature on residual learning.

Critically, the paper identifies a *domain gap* between masonry cracks (wide, structurally

controlled fractures in regular stone or brick patterns) and road surface cracks (thin, branching structures on uniform grey asphalt). Models trained on pavement data and evaluated on masonry show significant performance degradation, motivating dedicated masonry datasets and domain adaptation strategies. This finding is the primary motivation for the fine-tuning experiments in Tasks 5 and 1b of this project.

H. Heritage Building Damage Assessment

H.1. Computer Vision for Cultural Heritage: A Survey

Mishra and Lourenço [17] provide a comprehensive survey of deep learning and computer vision methods applied to damage detection in cultural heritage structures. The survey covers six damage categories: cracking, spalling, moisture damage, biological growth, efflorescence, and material loss. For each category, it reviews the available datasets, dominant architectures, and reported performance metrics, and identifies open research gaps.

Several findings are directly relevant to heritage damage assessment systems. First, the authors note that *no large-scale, multi-class, pixel-level annotated dataset* exists for heritage buildings comparable to COCO or ADE20K for general scenes. Available datasets contain at most a few hundred images per damage category, necessitating heavy use of transfer learning and data augmentation. Second, classification tasks (is damage present?) are substantially more tractable than localisation tasks (where is the damage?), with classification models consistently reaching >90% accuracy while detection and segmentation models typically achieve mAP or Dice below 60% on heritage-specific test splits.

Third, the survey identifies the *multi-class co-occurrence problem*: heritage buildings frequently exhibit multiple co-occurring damage types at different locations, requiring either multi-label classification or multi-class detection rather than the binary crack/intact formulation used in road-surface datasets. This limitation motivates the DACL10k multi-class detector (Task 9), which simultaneously localises crack, efflorescence, spalling, weathering, and wetspot damage.

Finally, the authors note that most studies evaluate on in-distribution test splits (same building type, similar lighting conditions) and rarely quantify cross-building or cross-country generalisation. This directly frames the central research question of the current project: whether OmniCrack30k-trained features can generalise to Albanian heritage building facades.

H.2. The DACL10k Dataset

The DACL10k (Damage Condition Labels for 10k images) dataset [16] provides polygon-annotated structural damage across 19 damage classes for approximately 10,000 images collected from civil engineering structures including bridges, facades, and retaining walls. The annotations cover both heritage and modern concrete structures, making it one of the most diverse structural damage datasets publicly available.

For the current project, DACL10k is used to train a 5-class multi-class detector (YOLOv8s) covering crack, efflorescence, spalling, weathering, and wetspot. The dataset’s polygon annotations are converted to bounding boxes for YOLO training, introducing annotation noise from the conversion (tight polygon contours become larger bounding boxes that

include background pixels). The resulting model achieves 13.5% mAP@50 overall, with weathering (22.7%) and spalling (16.2%) performing best due to their larger and more spatially consistent appearance compared to the thin, irregular morphology of cracks (12.7%) and wetspots (6.6%).

I. Transfer Learning and Domain Adaptation

I.1. Transfer Learning for Computer Vision

Pan and Yang [18] formalise the transfer learning problem: given a source domain $\mathcal{D}_S$ with abundant labelled data and a target domain $\mathcal{D}_T$ with scarce labelled data, the objective is to leverage knowledge learned in $\mathcal{D}_S$ to improve performance in $\mathcal{D}_T$. The survey distinguishes inductive, transductive, and unsupervised transfer scenarios based on whether labels are available in the source and target domains.

In the context of visual feature learning, transfer from ImageNet (1.2M images, 1000 classes) to a target visual task has become standard practice. Yosinski et al. [19] provide empirical evidence that the first convolutional layers of a CNN learn general edge and texture detectors that transfer almost universally, while later layers become progressively more domain-specific. This stratification motivates the common strategy of fine-tuning only the later layers while keeping early layers frozen, which is exactly the approach adopted in Tasks 5 and 1b.

For heritage crack detection, the transfer learning chain involves two levels of domain shift: (1) ImageNet photographs to crack images (general visual features to crack-specific textures), and (2) industrial crack images (OmniCrack30k road surfaces) to heritage masonry cracks. The first shift is well supported by the literature; the second is the primary challenge. The domain adaptation experiments in this project demonstrate that fine-tuning with only 250 heritage-labelled images increases mAP@50 by 168% relative (8.8% to 23.6%), consistent with the transfer learning literature’s finding that even small amounts of target-domain data substantially improve performance when the source-domain model provides good initialization.

The concept of *catastrophic forgetting* [23] is also relevant: fine-tuning all layers with a high learning rate can destroy source-domain features and reduce performance below the pretrained baseline. The project addresses this by freezing the first 10 backbone layers and using a learning rate of 10^{-3} ($10\times$ below the default YOLOv8 initial rate), consistent with standard fine-tuning practice in the transfer learning literature.

J. Visual Explainability

J.1. Grad-CAM: Gradient-weighted Class Activation Mapping

Selvaraju et al. [6] propose Gradient-weighted Class Activation Mapping (Grad-CAM), a technique for producing visual explanations of CNN decisions without architectural modifications. Given a target class c , the method computes the gradient of the class score y^c with respect to the feature map activations A^k of the final convolutional layer:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}, \quad (9)$$

where Z is the number of spatial positions. The Grad-CAM heatmap is obtained by computing a weighted combination of the feature maps and applying ReLU (to retain only positive contributions):

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right). \quad (10)$$

The ReLU ensures that only features that positively influence the class score are visualised, filtering out features that suppress the prediction. Grad-CAM is architecture-agnostic (applicable to any CNN) and class-discriminative (it produces different maps for different target classes). The resulting heatmap is upsampled to the input resolution for visualisation.

For heritage damage assessment, Grad-CAM provides a qualitative verification that the classifier has learned structurally meaningful features rather than dataset-specific shortcuts. Applied to EfficientNet-B4, Grad-CAM confirms that the model attends to crack edges and branching structures in both in-distribution (HistoricalCrack) and out-of-distribution (CrackForest, Masonry) images, providing evidence that the learned representations are semantically meaningful and generalise across visual domains.

J.2. Grad-CAM++: Sharper and More Localised Explanations

Chattopadhyay et al. [11] identify a limitation of standard Grad-CAM: for images containing multiple instances of the target class, the heatmap tends to produce a single coarse activation region covering all instances rather than localised maps for each. They propose Grad-CAM++, which replaces the uniform spatial average in the weight computation with a position-specific weighting:

$$\alpha_k^c = \sum_{i,j} w_{ij}^{kc} \cdot \text{ReLU}\left(\frac{\partial y^c}{\partial A_{ij}^k}\right), \quad (11)$$

where the weights w_{ij}^{kc} are derived from higher-order gradients (second and third partial derivatives), giving larger weights to spatial positions where the feature map activation contributes most to the positive gradient. This yields sharper, more localised maps that are particularly valuable when multiple crack instances are present in a single facade image.

In the failure analysis (Task 6), Grad-CAM++ is applied to EfficientNet-B4’s final MBConv block. The sharper localisation enables precise attribution of false positive predictions to specific image regions: intact images misclassified as cracked show Grad-CAM++ attention on mortar joints, shadow edges, and surface staining features that are visually similar to crack boundaries at the resolution of the final feature map. This analysis directly informs the domain adaptation strategy in Task 1b.

K. Image Quality, Structural Similarity, and Inference Efficiency

K.1. Structural Similarity Index (SSIM)

Wang et al. [20] propose the Structural Similarity Index (SSIM) as an alternative to pixel-level image quality metrics (MSE, PSNR) that better aligns with human visual perception. Their key insight is that the human visual system is adapted to extract structural information from scenes, so image quality degradation should be measured in terms of the loss of structural content rather than absolute pixel differences. SSIM

computes a local similarity score combining three components: luminance l , contrast c , and structure s :

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (12)$$

where μ_x, μ_y are local means, σ_x^2, σ_y^2 are local variances, σ_{xy} is the local cross-covariance, and C_1, C_2 are stability constants. The SSIM map is computed over sliding windows and combined into a single global score in $[-1, 1]$.

For heritage damage monitoring, SSIM is particularly valuable as a change detection metric: comparing photographs of the same building facade taken at different dates, the SSIM map spatially localises regions of structural change (new cracks, material loss, staining) as low-SSIM regions. The complementarity of SSIM with deep learning segmentation is a key design principle of the Before/After comparison mode in the web application: SSIM provides a model-agnostic change signal that does not depend on the segmentation model’s ability to detect every damage type.

In the synthetic degradation analysis (Task 8), SSIM drops monotonically from 1.000 (Stage 0, intact) to 0.321 (Stage 3, severe), with the SSIM change map correctly localising crack and staining regions as the primary sources of structural dissimilarity.

K.2. SAHI: Slicing Aided Hyper Inference

Akyon et al. [21] address the challenge of detecting small objects in high-resolution images with a standard-resolution detector. A detector trained on images resized to a fixed resolution (e.g., 640×640) is unable to resolve objects whose bounding boxes span fewer than ~ 20 pixels at that resolution, a problem common in aerial and satellite imagery, microscopy, and industrial inspection. Their Slicing Aided Hyper Inference (SAHI) framework tiles the input image into overlapping patches at the detector’s native resolution, runs inference on each patch, and merges the resulting detections back to original image coordinates using Non-Maximum Suppression.

The SAHI framework is detector-agnostic and requires no retraining: it is a post-hoc inference strategy. On VisDrone and COCO-based aerial benchmarks, SAHI improves small object AP by +6 to +10 points relative to standard inference.

Applied to the heritage crack detection problem (Task 7 in the earlier evaluation), SAHI is evaluated on the hypothesis that heritage masonry cracks, which are thin and locally fine-grained, might benefit from tile-based inference. The experimental result contradicts this hypothesis: SAHI underperforms standard inference (18.9% vs. 28.0% mAP@50). The explanation is that heritage masonry cracks are not genuinely *small objects* in the SAHI sense: they typically span a significant fraction of the bounding box at standard 640×640 resolution. Tile-based inference instead introduces patch boundary artifacts, loss of global crack context, and lower detector confidence per tile, reducing overall mAP. This negative result is an informative finding: SAHI is domain-specific and should not be applied without empirical validation.

L. Synthesis and Implications for Heritage Assessment

The surveyed literature collectively establishes the technical foundation for a multi-task heritage damage assessment pipeline. Several cross-cutting themes emerge.

Residual architectures are universal. From ResNet-34 (segmentation encoder) to EfficientNet-B4 (classifier) to the CSP-ResNet backbone of YOLOv8, every component of the pipeline is built on residual connections. This reflects the stability and gradient-flow benefits documented by He et al., which are especially important for fine-tuning scenarios where the learning signal from a small target-domain dataset is insufficient to train deep networks from scratch.

Multi-scale representations are necessary but not sufficient. FPN, ASPP, U-Net skip connections, and FPHBN all address the multi-scale challenge of crack detection. However, the SAHI experiment demonstrates that scale-invariance must come from the architecture’s feature representation, not from test-time tiling. The failure of SAHI on heritage cracks, despite its success on aerial small-object detection, illustrates that design choices must be empirically validated in the target domain rather than adopted from adjacent applications.

Domain adaptation is the central bottleneck. The literature on transfer learning (Pan and Yang; Yosinski et al.) establishes that early features transfer but later features are domain-specific. The cross-domain evaluation results confirm this: the OmniCrack-trained detector collapses completely on heritage masonry (mAP@50: 1.1%), while the ImageNet-pretrained classifier generalises remarkably well (recall: 76–90%). The difference reflects the relative specificity of the tasks: binary crack/intact classification exploits generic texture features, while bounding-box regression requires domain-specific spatial priors that do not transfer across surface types.

Explainability bridges performance and trustworthiness. Selvaraju et al.’s Grad-CAM and Chattopadhyay et al.’s Grad-CAM++ are not optional additions to the pipeline; they constitute the primary evidence that the models have learned the correct features rather than dataset shortcuts. In the heritage preservation context, where model decisions may inform costly conservation interventions, a system that attends to mortar joints and misclassifies them as cracks is not acceptable even if its aggregate accuracy is high. The 0% false negative rate of the classifier and the Grad-CAM evidence of crack-feature localisation together constitute a stronger argument for deployment than accuracy alone.

Heritage-specific data remains the critical gap. Mishra and Lourenço’s survey, Dais et al.’s masonry dataset, and the DACL10k benchmark collectively demonstrate that domain-specific annotated data is the primary bottleneck for heritage CV systems. The 250-image fine-tuning dataset used in this project produces meaningful improvements (+169% relative mAP) but falls far short of the performance achievable with purpose-built heritage crack datasets of comparable scale to OmniCrack30k. This motivates future work on semi-supervised learning, synthetic data generation (as explored in Task 8), and collaborative dataset collection across heritage institutions.

M. Article List

The following 20 articles form the reading basis of this survey. Authors are listed as in the original publications; conference and journal abbreviations follow standard CV convention.

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